

# Tunisia, audited — the complete audit

Five absences. Nine legs. One metabolism. Every number recomputed in code, every correction applied into the design, every claim graded.

Provenance law: MEASURED = cited anchor · DERIVED = arithmetic on measured values · ASSUMED = declared default. commercially-validated = 0 — no buyer has signed anything; published as zero on purpose. Generated 2026-09-10 by nations\_audit.py / make\_nations\_pdfs.py — the same code that renders the page.

## 1. The five absences and the one metabolism

Five structural absences — monetary sovereignty (debt \$40.5B, ~79% of GDP, 26.2% of export revenues to service), sovereign compute (InstaDeep: founded in Tunis, sold for \$680M), strategic materials (phosphogypsum at Gabès costing the sea >\$58M/yr), water, nutrition — answered by nine cash-positive legs on one metabolism of power, water, data and workforce. The audit is finished; the corrections are in the design.

## 2. The FX bridge — every dollar, by leg

Each leg below is volume × price × utilization, with the double-counting checks applied — the design carries the checked ledger.

| LEG                                           | TYPE                | VOLUME                                            | PRICE                                                                                                                                          | \$M/YR    | GRADE       |
|-----------------------------------------------|---------------------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------|
| Strategic metals (U + heavy REE, acid stream) | gross export        | 716,500 lb U3O8 + 585 t REO                       | \$80-95/lb U3O8 · \$30/kg REO                                                                                                                  | \$75-86M  | STRONG      |
| Solar fuel substitution (500 MW)              | import substitution | 745 GWh/yr displaced (876 GWh less 131 GWh to H2) | \$142/MWh (gas \$12/MMBtu x 11 MMBtu/MWh + \$10/MWh O&M); government anchor: 1,100 GWh ≈ \$125M/yr gas saving for the same 500 MW fleet (Pg14) | \$95-116M | PLAUSIBLE   |
| Hydrogen → ammonia (loop-internal offtake)    | import substitution | 2.63 kt H2/yr → 14.9 kt NH3 for the GCT chain     | \$500/t imported ammonia displaced (external offtake upside: \$8-9.5/kg H2)                                                                    | \$7-25M   | SPECULATIVE |
| Terfas (desert truffles)                      | gross export        | 940 t/yr mid (2,500 ha x 376 kg/ha; 70% exported) | EUR20-40/kg farm-gate                                                                                                                          | \$14-29M  | PLAUSIBLE   |
| Food import substitution                      | import substitution | 1% of the \$3.0B food import bill                 | imported protein \$15-25/kg vs local \$5-10/kg                                                                                                 | \$30-30M  | SPECULATIVE |

| LEG                                        | TYPE                     | VOLUME                                        | PRICE                                | \$M/YR     | GRADE                                                                                                   |
|--------------------------------------------|--------------------------|-----------------------------------------------|--------------------------------------|------------|---------------------------------------------------------------------------------------------------------|
|                                            |                          | (frame; volume bridge is a pilot deliverable) |                                      |            |                                                                                                         |
| Compute colocation services                | gross export (services)  | colo park, tenant-dependent                   | market colocation rates              | \$15-15M   | <i>SPECULATIVE</i>                                                                                      |
| ESG financing delta                        | financing delta (not FX) | 100-150bp on ~\$800M portfolio debt           | interest saved, not foreign exchange | \$8-12M    | <b>PLAUSIBLE</b>                                                                                        |
| <b>Total external-account effect (mid)</b> |                          |                                               |                                      | ~\$279M/yr | <b>\$80M strong ·</b><br><b>\$137M plausible ·</b><br><b>\$61M speculative ·</b><br><b>\$0 verified</b> |

**By kind:** exports \$117M · import substitution \$152M · financing delta \$10M (interest saved, honestly labeled as not-FX). **By grade:** \$80M strong · \$137M plausible · \$61M speculative · \$0 verified — nothing is commercially validated yet, and that ledger reads zero on purpose.

### 3. The energy ledger — every kWh, by leg

| LEG                             | GWH/YR                              | THE FORMULA                                                                                 |
|---------------------------------|-------------------------------------|---------------------------------------------------------------------------------------------|
| Pump-as-turbines (return flows) | 3.9                                 | $70\% \times 150\text{k m}^3/\text{d} \times 50\text{ m} \times 75\% \times 8,760\text{ h}$ |
| Heat (kiln + exotherm)          | 30                                  | 9% of 333 GWh envelope                                                                      |
| Flexibility (load shift)        | 20                                  | dispatch value — not recovered joules                                                       |
| CO <sub>2</sub> ; mineralized   | 30 kt                               | mass, not energy                                                                            |
| <b>Total</b>                    | <b>33.9 recovered + 20 dispatch</b> | <b>isobaric ERD inside the 4 kWh/m<sup>3</sup> benchmark — never double-counted</b>         |

### 4. Phase 0 — every dollar, itemized

| ITEM                                                                                           | COST     | WHAT IT BUYS                       |
|------------------------------------------------------------------------------------------------|----------|------------------------------------|
| Water panel: 100 points, full suite (F, Cd, Pb, Cr, As, salinity, nitrate)                     | \$35,000 | the first anchor measurement       |
| Acid-stream assays: 12 points (U, REE, Ra-226, Po-210, full metals) + GCT access               | \$40,000 | the second anchor measurement      |
| Emissions inventory: 30 passive samplers (PM <sub>2.5</sub> /SO <sub>2</sub> /fluorides) + lab | \$25,000 | baseline for the health scoreboard |
| Cohort protocol: 3,000-family registry, ethics, data platform v0                               | \$35,000 | the DALY ledger becomes MEASURED   |

| ITEM                                                                                         | COST             | WHAT IT BUYS                                                                                                                                   |
|----------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Health pilot first tranche: 100 households + 2 schools, 90 days                              | \$58,000         | filters, clean-air boxes, KPIs                                                                                                                 |
| Legal + finance: health fund, industrial SPV, land-lease pre-agreements, tender registration | \$25,000         | structure before scale                                                                                                                         |
| Engineering + project management                                                             | \$35,000         | the 90 days are run                                                                                                                            |
| <b>List price</b>                                                                            | <b>\$253,000</b> | <b>list price \$253k; the two anchor measurements cost \$75k and land first — the claimed range is the phased-execution plan, not the list</b> |

## 5. Every claim, graded

| CLAIM                                                        | GRADE            | WHY                                                                                                                                                              |
|--------------------------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| \$40.5B debt (~79% GDP); 26.2% of export revenues to service | <b>STRONG</b>    | MEASURED via World Bank API on 2026-09-09: debt \$40.46B, debt/GDP 78.7%, service 26.17% of export revenues (2024)                                               |
| \$3.0B food and \$4.9B fuel imports                          | <b>STRONG</b>    | fuel \$4.895B verified on WITS/Comtrade (HS27, 2024); food per the WB food classification                                                                        |
| InstaDeep founded in Tunis, sold for \$680M                  | <b>STRONG</b>    | public deal record; the emblem for talent without infrastructure                                                                                                 |
| Gabès: >\$58M/yr of REE value lost to the sea                | <b>STRONG</b>    | Science of the Total Environment (2021): up to 1,523.67 t/yr REE, ~\$58M/yr losses                                                                               |
| U + heavy REE from the acid stream, ~\$80M/yr                | <b>STRONG</b>    | flowsheet industrial since 1978 (Freeport), patent expired, Morocco building it now; volume derived from cited primitives; Gafsa-specific assays are Phase 0     |
| ~\$285M/yr external-account effect, tiered                   | TIERED           | \$80M strong (proven flowsheet/cited anchor) · \$137M plausible (gated) · \$68M speculative (assumed volume/price)                                               |
| 33.9 GWh/yr recovered + 20 GWh/yr dispatch value             | DERIVED          | every leg computed from cited primitives; dispatch value honestly labeled — not joules                                                                           |
| \$2,592 per healthy year (water rung)                        | <b>PLAUSIBLE</b> | arithmetic DERIVED in code; DALY anchors transferred from Iranian studies — the cohort baseline makes it MEASURED                                                |
| Phase 0 = \$150-250k, itemized                               | <b>PLAUSIBLE</b> | itemized list price \$1.5M for the complete pilot — measurements, season, prototype, IP, team, 20% margin; the two anchor measurements cost \$75k and land first |
| Commercially validated                                       | <b>ZERO</b>      | no buyer has signed anything — published as zero on purpose                                                                                                      |

## 6. The hidden dependencies

| DEPENDENCY         | TYPE       | STATUS    | NOTE                                                      |
|--------------------|------------|-----------|-----------------------------------------------------------|
| Desal brine permit | regulatory | in design | Gabes outfall / diffuser; permit required before SWRO FID |

| DEPENDENCY                         | TYPE           | STATUS        | NOTE                                                                              |
|------------------------------------|----------------|---------------|-----------------------------------------------------------------------------------|
| PG chemistry (Ra-226 partitioning) | technical      | Phase 0 assay | flowsheet keeps radium in gypsum (Freeport precedent); re-verify on Gafsa stream  |
| Uranium handling                   | regulatory     | gate          | CNSTN licensing, NORM transport, export controls                                  |
| REE separation capex               | capital        | later phase   | +\$244M solvent-extraction plant beyond the ~\$100M circuit                       |
| Solar bankability                  | financial      | gate          | DSCR ~0.9-1.0 at the record-low tariff; bankable region or grant support required |
| Grid connection                    | infrastructure | later phase   | 3 GW connection + 400 kV upgrades ~\$450M                                         |
| Compute accelerators               | regulatory     | gate          | export-control licenses; colocation until an anchor tenant is licensed            |
| Truffle yield                      | agronomic      | gate          | 376 kg/ha is the 20-yr Murcia average; gate = 3-season record $\geq 200$ kg/ha    |
| Export prices                      | market         | exposure      | U spot \$80-95/lb; truffle EUR20-40/kg — no contracted offtake yet                |
| Financing                          | financial      | uncommitted   | DFI channel modeled for solar/desal; nothing committed                            |

## 7. The structure and the pilot/capital separation

**The design is six independently bankable modules sharing infrastructure.** The loop is the outcome, never the precondition — no module's economics depends on another module succeeding, and every gate is public. World Bank CCDD (2023): Tunisia's water-sector investment need ~\$17.1B and energy-sector ~\$34.7B through 2050. The page is not that program and does not claim to be: Phase 0 is \$150-250k of measurements, and the whole core build is ~\$250-650M phased over years 2-5, module by module, each gated. The CCDD numbers confirm the scale of the problem; they do not compete with a pilot gate.

Africa Energy Portal (2026): 500 MW of utility-scale solar approved (~\$400M investment). Same class as the page's P3 module (500 MW) — the record-low ~\$31/MWh awards are the reference the DSCR gate is tested against.

## 8. The Fed transmission — measured, not restated

World Bank API, measured 2026-09-09 (Pg1): external debt **\$40.46B, 78.7% of GDP**; debt service **26.2% of export revenues**; reserves **\$9.34B** = 4.3 months of imports; food \$3.00B + fuel \$4.90B of imports; CA -1.5% of GDP; growth 2.5%, inflation 5.2% (2025).

**Channel 1 — the debt trap:** the “original sin” (Pg12) — no long-term borrowing in the dinar abroad — exposes the state to the Fed cycle (Pg13): Fed hikes lift global yields, refinancing turns expensive, and the 26.2% of export revenues that debt service absorbs compounds while reserves (4.3 months) drain.

**Channel 2 — imported inflation:** the rate differential pulls capital to US safe assets, the dinar depreciates, and the dollar invoice on food (\$3.00B) and fuel (\$4.90B) inflates. The BCT — autonomous, non-monetizing under Organic Law 2016-35 (Pg4/Pg5) — must hold rates to defend the dinar: correct orthodoxy, strangled growth.

**The loop’s answer:** owned export engines earn ~\$117M/yr of gross exports outside the Fed cycle; the substitution legs take \$152M/yr out of the dollar invoice directly; reserves rise, the rating runs B-/Caa1 → BB-, and the refinancing premium shrinks for the state with its own hard-currency engines. Not defiance of the Fed — independence from the channel.

## The household ledger — what this does to the cost of living

A state program that does not make life cheaper for its people is decoration. Each line, computed from the same primitives as the rest of this document; the quarterly price index in Gafsa and Gabès is the published enforcement: **the price of protein and vegetables must bend downward within 24 months of the first harvest.**

| HOUSEHOLD LINE | TODAY                                        | AFTER THE LOOP                                               |
|----------------|----------------------------------------------|--------------------------------------------------------------|
| Water          | \$1.00-1.50/m <sup>3</sup> (SONEDE produced) | \$0.35/m <sup>3</sup> coastal · \$0.34/m <sup>3</sup> inland |
| Protein        | \$15-25/kg imported                          | \$5-10/kg local                                              |
| Terfas         | export €20-40/kg                             | one third stays home at ≤€12/kg, contractual                 |
| Health         | the disease's bill                           | \$2,592 per healthy year vs the \$4,000/DALY standard        |
| Energy (grid)  | \$0.110/kWh industrial                       | \$0.031/kWh for the loop's loads; tariffs stay state policy  |
| Work           | 28% unemployment (54% women graduates)       | workshop, nursery, ponds, plants, crews                      |

## The energy ledger — what power costs, per party

Anchors cross-checked: Rw10 (0.352 TND/kWh industrial, 19 TWh), Pg14 (the government’s own 1,100 GWh = 5% of production, ~\$125M/yr gas saving), Rc13 (\$31/MWh inside the \$24-35 historical bid band), Pg15 (fuel imports \$4.895B, WITS-verified).

| PARTY                 | TODAY                                     | AFTER THE LOOP                            | THE DELTA                                               |
|-----------------------|-------------------------------------------|-------------------------------------------|---------------------------------------------------------|
| The loop's industries | grid \$0.110/kWh                          | <b>\$0.031/kWh</b> on the loop's PPA      | −72%                                                    |
| The state             | 19-22 TWh/yr, ~95% gas, \$4.90B fuel bill | 876 GWh/yr solar (4.0-4.6% of production) | −\$106-125M/yr gas imports (\$114-142/MWh band)         |
| The household         | subsidized STEG tariff                    | the cost stack beneath the tariff falls   | the inflation channel throttled — every dinar protected |

## The frontier — where the loop reaches, and where it honestly stops

The complete household budget, swept — nothing hidden, nothing promised beyond the frontier.

| HOUSEHOLD LINE                 | STATUS          | THE HONEST NOTE                                                                                                                    |
|--------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------|
| Electricity (household tariff) | <b>indirect</b> | household tariffs stay subsidized STEG policy; the loop lowers the cost stack they sit on and the subsidy burden the state carries |

| HOUSEHOLD LINE             | STATUS                    | THE HONEST NOTE                                                                                       |
|----------------------------|---------------------------|-------------------------------------------------------------------------------------------------------|
| Cooking gas (the bonbonne) | <b>out of scope</b>       | the loop displaces grid gas, not bottled LPG; the bonbonne subsidy is a separate state decision       |
| Transport fuel             | <b>out of scope today</b> | the hydrogen leg serves ammonia for the fertilizer chain; transport fuels are a later policy question |
| Wheat and bread subsidy    | <b>out of scope</b>       | the protein stack complements the grain program; it does not replace it                               |
| Water                      | <b>covered</b>            | cost floor \$1.00-1.50 -> \$0.35 coastal / \$0.34 inland                                              |
| Health                     | <b>covered</b>            | \$2,592 per healthy year vs the \$4,000 standard                                                      |
| Protein                    | <b>covered</b>            | \$5-10/kg local vs \$15-25/kg imported                                                                |
| Work                       | <b>covered</b>            | the workshop, nursery, ponds, plants, crews where unemployment is 28% (54% for women graduates)       |
| Housing                    | <b>out of scope</b>       | the loop does not build housing; named so it is not promised                                          |

## The water stack — line-by-line, with each credit's failure mode

The \$0.347/m<sup>3</sup> audit: the base is measured anchors; each credit is a loop-internal yield with a named failure mode; the degraded cascade still clears the \$0.68 gate.

| LINE                                                                                                                                                                                                                                                               | \$/M <sup>3</sup> | PROVENANCE                                                                                                                                      | WHAT MAKES IT FAIL                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Base — solar energy 4 kWh/m <sup>3</sup> at the loop's own PPA + capex + O&M                                                                                                                                                                                       | \$0.567           | MEASURED anchors (Rc13/Rw2/Rw3)                                                                                                                 | holds — if even the PPA lapses, grid power takes the base to \$0.88, still under SONEDE                                                                                                                                                                                                                                                                       |
| DC reject heat → RO feed preheat — one seawater pipe cools the servers, then arrives warm at the membranes (SEC −2.5%/°C)                                                                                                                                          | −\$0.002 net      | DERIVED — Leg 03 + Leg 01, same pipework                                                                                                        | the trade, stated: warmer feed saves ~\$0.007/m <sup>3</sup> of energy but accelerates biofouling + polyamide hydrolysis (~\$0.005/m <sup>3</sup> membrane debit) — the net is near zero, so the coupling's real justification is the DC's free cooling, not the water credit; lifetime extension comes from pretreatment, thermal stability and flux control |
| Kiln/exotherm preheat — the acid plant and board kilns warm the same feed                                                                                                                                                                                          | −\$0.000          | DERIVED — Leg 04/05 waste heat                                                                                                                  | negligible by design — carried at zero margin                                                                                                                                                                                                                                                                                                                 |
| Brine valorization — NaCl by solar-crystallization ponds (free sun); Mg(OH) <sub>2</sub> by lime precipitation from the loop's own kilns (134 t/yr lime, 0.14 GWh/yr — no evaporation), 100 kt to the board's fire retardant + 40 kt surplus sold at the park gate | −\$0.137          | DERIVED — process specified, CARMEn-anchored cost; the Mg(OH) <sub>2</sub> VALUE books in the board's own ledger, zero here until offtake signs | if the Mg(OH) <sub>2</sub> of-take is not signed by FID, the credit degrades to −\$0.03 (disposal avoided only) — the biggest single sensitivity                                                                                                                                                                                                              |

| LINE                                                                                                       | \$/M <sup>3</sup>            | PROVENANCE                                | WHAT MAKES IT FAIL                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------|------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shared civil works — one intake, one outfall, one grid connection serves RO + DC + metals in the same park | −\$0.032                     | DERIVED — 12% capex sharing               | if the park phases one module at a time, halve it to −\$0.016                                                                                                                                                                 |
| Dispatch value — the desal's own flexibility follows the loop's own solar curve                            | −\$0.015                     | DERIVED — 20 GWh shifted                  | fails to zero if the STEG contract does not reward flexibility — contractual, not hoped                                                                                                                                       |
| ESG financing delta — the health leg's public KPIs reprice the loop's own debt (100–150bp)                 | −\$0.026                     | DERIVED — nations_v3                      | fails to zero if lenders do not accept the KPI track record — bankable-region fallback keeps the module alive                                                                                                                 |
| Brine pump-as-turbine — the desal's own residual pressure                                                  | −\$0.002                     | DERIVED — 2.5 bar × 75%                   | fails to zero — immaterial                                                                                                                                                                                                    |
| <b>Integrated</b>                                                                                          | <b>\$0.352/m<sup>3</sup></b> | <b>28% of SONEDE's \$1.00–1.50 (Pg16)</b> | <b>the degraded cascade: if every soft credit fails, water still lands at \$0.521/m<sup>3</sup> — under SONEDE's own \$1.00–1.50, and the \$0.68 gate still clears with margin. The credits are the margin, not the plan.</b> |

## The Darwinian ledger — why business as usual loses

Protein built the human brain (Pg27); the same lever, applied on purpose, is the plan's deepest one. The business-as-usual ledger is the fear-facts: what “keep doing what we do” actually buys.

| FACT               | THE NUMBER                                       | WHAT IT MEANS FOR THE NEXT GENERATION'S BRAINS                              |
|--------------------|--------------------------------------------------|-----------------------------------------------------------------------------|
| Fluorosis belt     | 24% dental fluorosis (Rw1)                       | high-F water ↔ lower child IQ (Pg28) — BAU taxes the cortex before it forms |
| Protein poverty    | first 1,000 days (Pg29)                          | permanent cognitive deficit — the trees version, built child by child       |
| Brain drain        | tertiary emigration among world's highest (Pg30) | the country exports its cortex like its phosphate                           |
| The proof          | InstaDeep sold for \$680M                        | the cognitive gold was real — and sold, not kept                            |
| Debt               | 26.2% of export revenues to service (Pg13)       | the budget cannot buy the interventions the brain needs                     |
| The idle cohort    | 28% unemployment; 54% women graduates (Rm19)     | built brains, no environment — adaptation = exit                            |
| Inflation          | 5.15% on \$4.90B fuel + \$3.00B food             | the first cut is diet quality — the Fed reaches the children's IQ           |
| SELECTION PRESSURE | WITHOUT THE PLAN                                 | WITH THE PLAN                                                               |
| Water scarcity     | selects emigration — the skilled leave first     | the mesh re-specifies the environment; adaptation becomes staying           |
| Protein poverty    | caps cognition permanently                       | ponds + canteens + quota reverse the pressure                               |

| SELECTION PRESSURE | WITHOUT THE PLAN                                | WITH THE PLAN                                 |
|--------------------|-------------------------------------------------|-----------------------------------------------|
| The debt cycle     | the surplus is eaten; the economy cannot invest | owned exports + substitution cut the invoice  |
| Institutions       | gates stay closed; skills emigrate through them | public gates select for competence, in public |

## The open-field plan — terfas, hectare by hectare

| RUNG             | HECTARES | SEEDLINGS | PRODUCTION | VALUE                                    | OPENS ON                   |
|------------------|----------|-----------|------------|------------------------------------------|----------------------------|
| Gate (yrs 1-4)   | 100      | 600,000   | ≥200 kg/ha | first fruit ≥€20/kg                      | 3-season record + contract |
| Scale (yrs 4-8)  | 2,500    | 15M       | 940 t/yr   | €16544-29704M/yr                         | nursery output             |
| Ceiling (yrs 8+) | 3,000    | 18M       | 1,128 t/yr | inside 1,000-1,500 t/yr market cap (Rc8) | market offtake signed      |

6,000 mycorrhized seedlings/ha (Rc16) — the nursery is the first asset: 5M seedlings/yr of local-ecotype capacity, ~200 nursery + ~1,000 field + ~783 seasonal jobs at scale. Local wild genetics, not the imported soil biology that failed in Saudi Arabia (Rc17).

## The AI spine — the system's nervous system, and its boundary

The AI runs the measurement, the routing and the publication — never the decisions.

| COMPONENT           | WHAT IT DOES                                                                                                                                      | WHAT IT IS WORTH                                                    |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| The dispatcher      | routes every joule and m <sup>3</sup> in real time: solar → desal → electrolyzer → ponds → DC cooling, following the solar curve minute by minute | worth the 20 GWh/yr of dispatch value — the AI is what collects it  |
| The scoreboard      | computes and publishes every gate number automatically — filter uptime, urinary F, DSCR, yield records — tamper-evident, no human can edit a gate | the public contract, machine-audited                                |
| The price index     | tracks the Gafsa/Gabès household basket quarterly and publishes the bend — the people's enforcement metric, computed not curated                  | the number that proves the prices bend, or proves they do not       |
| The cohort engine   | runs the 3,000-family health baseline, dose-outcome analytics and anomaly flags for the clinical team                                             | the DALY ledger becomes a live dataset                              |
| The dispatcher twin | a digital twin of the loop (energy, water, brine, heat) that simulates before any physical change and audits after                                | the falsification ladder, automated                                 |
| The boundary        | the AI never sets a gate, never signs a contract, never prices a tariff — it measures, routes and publishes. The decisions stay human, in public. | the guardrail that keeps the nervous system a servant, not a master |

## The results framework — what a World Bank board reads first

The scoreboard IS the M&E system (PforR class, Pg23).



| INDICATOR                              | BASELINE                                 | TARGET                                       | DATA SOURCE            |
|----------------------------------------|------------------------------------------|----------------------------------------------|------------------------|
| Filter uptime, pilot households        | 0% (filters not installed)               | ≥70%                                         | scoreboard, live       |
| Urinary fluoride, pilot children       | baseline to be measured (Day 1-30)       | –20%                                         | cohort engine          |
| Household price index, Gafsa/Gabès     | baseline Qo                              | bend downward ≤24 months after first harvest | price index, quarterly |
| Water produced cost                    | SONEDE \$1.00-1.50/m <sup>3</sup> (Pg16) | ≤\$0.35/m <sup>3</sup> integrated            | plant telemetry        |
| Energy recovered inside the loop       | 0 GWh/yr                                 | 33.9 GWh/yr + 20 GWh dispatch                | dispatcher ledger      |
| Gas imports displaced                  | \$4.895B fuel bill (Pg15)                | –\$106-125M/yr                               | WITS Comtrade, annual  |
| Uranium recovered from the acid stream | 0 (lost to the sea)                      | ≥85% recovery gate                           | assay chain            |
| Terfas hectares planted                | 0 ha                                     | 100 ha gate → 2,500-3,000 ha                 | nursery registry       |
| Jobs in the basin                      | 28% unemployment (54% women graduates)   | workshop → nursery → plants → crews          | employment registry    |

## The time bomb, per party — every habit, every fuse

| THE PARTY                                     | THE HABIT                                                                                               | THE NUMBERS                                                                                                                           | THE FUSE                                                                                                                                                           |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The small farmer (diesel)                     | habits: irrigating thirsty crops from 15-60 m phreatic wells, then 80-250 m tubewells as the table fell | the shallow wells are already dead in the stressed cones (drawdown 30-40 m); the tubewell crosses the pipeline's flat price at ~119 m | fuse: ~5-15 years — the ladder takes him by money; the meter is his rescue, not his enemy                                                                          |
| The PV farmer                                 | habit: photovoltaic pumping at near-zero marginal cost — the deepest-drilling of the small holders      | survives the money rung entirely; the CT declines 0.5-1.5 m/yr under him and the quality degrades                                     | fuse: ~10-25 years — the salt and the empty well take him, exactly where the meter still saves him                                                                 |
| The cooperative (STEG)                        | habit: subsidized-tariff pumping for collective plots                                                   | crossing depth ~219 m; the same table falls under the same grid                                                                       | fuse: ~10-20 years — the subsidy dies before the well does; the flat price outlives both                                                                           |
| The bottled-water companies                   | habit: mining the fossil sea for export — CI wells 1,200-2,200 m, the deepest of all                    | the CI declines 1-3 m/yr (nodes to 5); recharge ~zero — every bottle is permanent depletion                                           | fuse: centuries of volume, but the endowment is non-renewable — the register must regulate the deepest drillers first, because their habit is the least reversible |
| The industries (CPG 300-750 m, GCT 150-400 m) | habit: industrial pumping for washing and process water                                                 | the Gabès coastal zone already reads 5 to >10 g/L; the saline front advances 15-100 m/yr                                              | fuse: ~5-15 years for the coastal wells — the loop's reuse leg replaces their pumping cheaper, before the salt does it for them                                    |

| THE PARTY                    | THE HABIT                                                                                                                              | THE NUMBERS                                                                                             | THE FUSE                                                                                                                                             |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| SONEDE                       | habit: serving the cities at ~\$1.50/m <sup>3</sup> from 200-600 m wells (ultra-deep 1,200-1,800 m at Kebili/Tozeur, water at 55-70°C) | the pipeline's \$0.35-0.52 undercuts the state's own production cost — the rational switch is immediate | fuse: the shortest — the state buys instead of pumping, and its wells retire into the strategic reserve                                              |
| The oases (Deglet Nour belt) | habit: the region's thirstiest crop, fed by the CT under the date palms                                                                | the date-pumping zones have already drawn 30-40 m; the CT falls 0.5-1.5 m/yr                            | fuse: ~15-25 years of productive water — the oasis leg's reuse + the pulse are the only thing standing between the palms and the salt                |
| The households               | habit: the tap — the most innocent and the most exposed                                                                                | the dams hit 20.5-22% fill in autumn 2023; Sidi Salem below 16% (~95-105 Mm <sup>3</sup> )              | fuse: the next 2023-class drought — rationing becomes catastrophe; the desal is their lifeboat, and the household ledger says it arrives 65% cheaper |
| The state (BCT, treasury)    | habit: importing the fuel to pump, the food to eat, and borrowing the difference                                                       | the FX bridge's \$278.70M/yr and the debt's 26.17% service ratio both hang on the water                 | fuse: each drought year tightens the trap — the loop's substitutions (fuel, food, FX) are the only defuser the treasury holds                        |

the clock, read per party: every habit has its fuse, and every fuse shortens with every metre of decline. The small farmer: 5-15 years. The PV farmer: 10-25. The cooperative: 10-20. The bottlers: centuries, but irreversible. The industries: 5-15 by salt. SONED: the switch is already economic. The oases: 15-25. The households: the next drought. The state: every drought year. The project is not terraformation — it is defusing, party by party, habit by habit, fuse by fuse — and the meter is the hand that starts cutting wires.

## The verified aquifer facts — every number cited, nothing believed

| THE FACT                  | THE NUMBER                                                                                                                                                                                                                      | WHAT IT MEANS FOR THE STUDY                                                                                                                    | SOURCE                           |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| The national deficit      | abstraction 2,700-2,850 Mm <sup>3</sup> /yr against ~2,150 Mm <sup>3</sup> /yr of renewable recharge — a deficit of 550-700 Mm <sup>3</sup> /yr; deep fossil aquifers mined at 130-150% of recharge                             | the pulse's 8-14 Mm <sup>3</sup> is ~1.5-2.5% of the hole — the loop ARRESTS the south's stressed zones; the full program is the national cure | OSS / DGRE / FAO AQUASTAT (Pg41) |
| The decline, measured     | Complexe Terminal 0.5-1.5 m/yr since the late 1970s (cones >2 m/yr, 30-40 m total drawdown in the date zones — the artesian springs are gone); Continental Intercalaire 1-3 m/yr, up to 5 m/yr at the Tozeur/Kebili/Gafsa nodes | the money clock accelerates: the ladder's crossing depths arrive sooner than the earlier 0.5-1 m band implied                                  | OSS SASS Phase I-II (Pg41)       |
| The bottles drill deepest | southern bottled waters tap the same aquifers: CT boreholes 80-250 m, CI wells 1,200-2,200 m; Sabrine's primary source is Aïn Octor (Kairouan-Sbikha), the southern lines use local CT/CI boreholes                             | the bottle and the farmer's well drink the same fossil bank — the country exports its strategic endowment in plastic                           | ONTH / DGRE cadastres (Pg41)     |

| THE FACT            | THE NUMBER                                                                                                                                                                                                        | WHAT IT MEANS FOR THE STUDY                                                                              | SOURCE                                        |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| The parties' depths | farmers 15-60 m phreatic + 80-250 m tubewells; SONEDE 200-600 m (geothermal CI wells 1,200-1,800 m at Kebili/Tozeur, water arriving at 55-70°C); CPG 300-750 m; GCT 150-400 m                                     | corrects the ladder's party notes: CPG drills deepest of the industries, the state deepest of all        | DGRE / CPG audits / SONEDE reports (Pg41)     |
| The dams            | total 2,350-2,550 Mm <sup>3</sup> across 36-37 dams (minus ~20% siltation); record low 20.5-22% in autumn 2023; Sidi Salem 580 Mm <sup>3</sup> usable (below 16% in mid-2023); Sidi el Barrak 286 Mm <sup>3</sup> | confirms the fill-them-all bank: ~1,800 Mm <sup>3</sup> of empty, paid-for volume stands in the register | DGBGTH / ONAGRI (Pg42)                        |
| The salt clock      | the Jeffara's piezometric cone sits 5-10 m below sea level; the marine front advances 50-100 m/yr; coastal salinity 5-8+ g/L, the interface within 30-60 m depth and 2-5 km of the shore                          | the ladder's QUALITY rung has its measured speed: 50-100 m/yr toward the wells                           | UNESCO-IHP / J. African Earth Sciences (Pg42) |
| The regulatory rung | irrigation is 65-75% groundwater nationally, >95% in the south; illegal boreholes 21,000-25,000+ officially estimated, 35,000+ by the unions                                                                      | the register's battlefield is real and counted — the meter and the register fight a numbered war         | MARHP / DGRE / FAO Water Report 45 (Pg42)     |

every number above survived the house stack: dimensionally clean, internally consistent, cited to named institutions — and it corrected four earlier bands (the deficit scale, the CI decline, the CPG/CI depths, the dam capacities). The study now carries the corrected figures as the design: the pulse is the south's arrest, the substitution is the cure, the ladder is the enforcement, and the verified numbers are the map.

## The ladder of extinction — who stops drilling, and at what depth

| THE DRILLER                     | PUMP | ENERGY      | D*     | THE NOTE                                                                                                                                                                                                                                                                    |
|---------------------------------|------|-------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The small farmer (diesel)       | 40%  | \$0.300/kWh | ~119 m | diesel pumps at ~40% efficiency — the thinnest wallet, the first to cross                                                                                                                                                                                                   |
| The PV farmer (photovoltaic)    | 55%  | \$0.015/kWh | ~253 m | the owner's own finding: PV pumping costs almost nothing at the margin — the energy rung nearly vanishes, and the farmer survives economically to much greater depth                                                                                                        |
| The STEG-subsidized cooperative | 55%  | \$0.070/kWh | ~219 m | grid electricity at the subsidized agricultural tariff — deeper pockets, later crossing                                                                                                                                                                                     |
| The stolen-energy driller       | 55%  | \$0.000/kWh | ~264 m | illegal connections make the marginal cost zero — here the economic rung fails entirely, and only the REGULATORY rung holds: the wells register, the abstraction permits, the published metering                                                                            |
| The industrial wells (CPG, GCT) | 75%  | \$0.090/kWh | ~221 m | industrial pumps, negotiated energy — but the brackish cap arrives BEFORE the money: the deep southern aquifer turns saline, and the loop's reuse leg sells them process water cheaper than pumping                                                                         |
| SONEDE's own wells              | 70%  | \$0.110/kWh | ~210 m | the state's strongest driller — and the one with the clearest math: SONEDE already spends ~\$1.50/m <sup>3</sup> produced; the pipeline's \$0.35-0.52 undercuts the state itself, so the rational move is to BUY, stop pumping, and hold the wells as the strategic reserve |

$C(d) = \text{drill\_cost} \times d \div \text{well\_life} \div \text{annual\_volume} + (\rho \cdot g \cdot d \div 3.6e6 \div \eta) \times \text{energy\_price} + \text{O\&M}$  — each party's full cost per m<sup>3</sup>, rising with every metre the table falls. The pipeline's price is flat: it is the ONLY player off the ladder.

but the ladder has THREE rungs, and money is only one of them. Where PV or stolen energy make the marginal cost near zero, the ENERGY rung fails — and the other two rungs bind instead: (1) the QUALITY rung — the deep southern aquifer turns brackish, and no pump can unsalt it at the field; the desal's water is the best quality on the map; (2) the YIELD rung — as the table falls, every well's yield collapses (the same capex, less water, until the well goes dry). The pipeline beats all three: cheapest where energy is paid, best quality everywhere, unlimited yield. And where energy is stolen, the REGULATORY rung holds: the wells register, the abstraction permits, the electronic meter that never forgets.

**the sequence: the diesel farmer crosses at ~119 m, the PV farmer survives economically far deeper but dies on QUALITY and YIELD as the table falls, the cooperative crosses at ~219 m, the industrial wells meet the brackish cap and buy the reuse instead, and SONEDE's wells retire last into the strategic reserve — a ladder of extinction where each rung switches because one of the three limits — money, salt, or an empty well — finally arrives.**

the electronic meter: the farmer, the cooperative and the industry become customers — the pipeline delivers, the meter counts, the price is published, and the wells register records every well that stops. The remaining levels become the nation's strategic reserve: the level becomes ours, exactly as the owner said.

the ladder's verdict, computed honestly against the DEGRADED pipeline price (\$0.52, the conservative test): the diesel farmer's well stops paying at ~119 m; the PV farmer — the owner's key insight — outlives the energy rung and must be taken by QUALITY and YIELD as the table falls; the cooperative crosses at ~219 m; CPG/GCT's wells die of salt before money and buy the reuse instead; and SONEDE — already paying ~\$1.50/m<sup>3</sup> to produce what the pipeline sells for \$0.35 — rests its wells last, holding them as the strategic reserve. Three rungs — money, salt, emptiness — and the pipeline is off all three. Every switch is a well that rests and a meter that pays; where energy is stolen, the register enforces what the price cannot. The aquifer war ends with a flat price no driller can beat, water no salt can touch, and a meter that never forgets.

**The two faces of the sovereign river — every gift, every cost, side by side**

| THE GIFT             | THE NUMBER                                                                              | THE NOTE                                                      |
|----------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------|
| The aquifer refilled | 8-14 Mm <sup>3</sup> /yr of silent recharge through the oued beds' alluvium             | the KPI measures it — drought insurance with zero evaporation |
| The banks bloom      | the seed banks wake on the pulse (Junk 1989)                                            | flow, then bloom — the ecosystems the wadis evolved for       |
| The clouds invited   | ~3 Mm <sup>3</sup> /yr of vapor into the recycling band (10-30% re-precipitation, Pg44) | booked low; the rain gauges are the referee                   |
| The dams refilled    | ~1,800 Mm <sup>3</sup> of empty paid-for volume, restored over decades                  | the nation's water bank AND ~956 GWh of gravity reserve       |
| Sovereignty          | the source is the sea; the release points are national                                  | the upstream dams are not disputed — they are made irrelevant |
| The rivers return    | the oueds flow again — the landscape's original map                                     | the geography the Romans' aqueducts and the qsours followed   |

| THE COST                  | THE NUMBER                                                                                                                    | THE CONTROL                                                                                                                                          |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| The allocation            | the pulse takes 20 Mm <sup>3</sup> /yr = 36% of the desal output — the largest single environmental commitment in the program | control: it starts at the ecological minimum and scales ONLY with the pilot's measurements                                                           |
| The energy                | 25.6 GWh/yr for the pulse; 71 GWh/yr (8% of the solar farm) at the full program                                               | control: priced in the water stack; the solar farm carries it without new capex                                                                      |
| The vector risk           | standing water in the wadis can breed mosquitoes — the health leg's own frontier                                              | control: pulse design keeps the flow MOVING (no stagnant pools); the release calendar avoids the breeding season's peaks; the health cohort monitors |
| The mismanagement risk    | a mistimed pulse can scour banks or flood downstream plots                                                                    | control: the release calendar is the MPC dispatcher's job — published, gated, reversible in hours                                                    |
| The recycling uncertainty | the 10-30% re-precipitation band is continental-class; the local return may be lower                                          | control: booked at zero revenue; the pilot oued measures downwind humidity before scaling                                                            |
| The shared basin          | the recharge benefits a transboundary aquifer system — a legal, not technical, dimension                                      | control: framed as cooperation; the measured recharge is published — what we give the shared basin is public, not hidden                             |

the two faces, side by side, with their numbers: the sovereign river is the largest single environmental allocation in the program (36% of the desal output) and the study says so on the same line as its benefits. Every risk has a named control — the gate, the calendar, the moving water, the published measurements — and the falsification is cheap: one pilot season decides. The page hides neither face; a project that hides its costs has no right to claim its gifts.

## The sovereign river — the bulletproof ledger (defend or kill)

| THE ATTACK                     | THE ARITHMETIC                                                                                                                                                                                                       | THE VERDICT                                                                                                                                                                               |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The energy kill-shot           | lifting water 400 m costs 1.28 kWh/m <sup>3</sup> — the 20 Mm <sup>3</sup> pulse needs 25.6 GWh/yr, the full 55 Mm <sup>3</sup> program 71 GWh/yr = 8% of the 876 GWh solar farm                                     | already inside the water stack's \$0.352/m <sup>3</sup> (the corridor's pumping is a priced line) — no hidden energy debt                                                                 |
| The water-accounting kill-shot | 40-70% of the released flow infiltrates — that is not a loss, it is the aquifer recharge, and the scoreboard's aquifer-level KPI measures it; ~15% becomes vapor — the recycling band's input, booked at the low end | if the pilot oued measures infiltration below 40% and no seed-bank response, the pulse dies at the gate — the falsification is built in, and the release starts at the ecological minimum |
| The sovereignty kill-shot      | the release points are Tunisia's own dams and the loop's own corridor; the source is the sea inside the national zone — upstream decisions cannot reach either                                                       | the loop does not dispute the upstream dams; it makes them irrelevant — the country's first rivers that no neighbour can turn off                                                         |
| The cost kill-shot             | the pulse claims ZERO revenue — it is an environmental expenditure of ~\$1-2M/yr of pumping energy, inside the water stack's opex                                                                                    | nothing to attack: no revenue claimed, no subsidy required, the DSCR and the MIGA wrap are untouched — the pulse is a gift to the land, not a bet on it                                   |

verdict: DEFEND — with the ledger closed. The sovereign river costs 1.28 kWh/m<sup>3</sup> at 400 m, ~8% of the solar farm, already priced inside the water; its infiltration is the aquifer's recharge (measured), its vapor is

the recycling band's input (booked low, rain-gauge-gated), and its sovereignty is real because both the release points and the source are national. What the pulse does NOT claim is exactly what cannot be attacked: no revenue, no subsidy, no change to the capital stack. And its killer test costs a season: one pilot oued, piezometers and gauges that already exist, and the numbers decide. The idea survives because it is small, measurable, and irreversible only where it should be — in the aquifer, not on the balance sheet.

## The oued pulse — the country's rivers, reborn from the sea

| THE JOB                               | THE MECHANISM                                                                                                                                          | THE NOTE                                                                                                                                                                                                                                              |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The aquifer, refilled silently        | 40-70% of the released flow infiltrates through the oued beds' coarse alluvium — 8-14 Mm <sup>3</sup> /yr of groundwater recharge                      | subsurface storage is the country's best drought insurance: no evaporation, no silt, and it lifts the aquifer-recovery KPI already on the scoreboard                                                                                                  |
| The banks, woken                      | the seed banks of the oued beds respond to the pulse — flow, then bloom                                                                                | the flood-pulse concept (Junk 1989): the ecosystems of dry riverbeds evolved AROUND the pulse; a regular release is not artificial — it is the returning heartbeat                                                                                    |
| The clouds, invited back              | ~3 Mm <sup>3</sup> /yr rises as vapor from the wetted beds and the revived vegetation                                                                  | the recycling band (10-30% re-precipitation downwind, Pg44) turns the release into clouds over the steppe — the mist's promise, at landscape scale                                                                                                    |
| The landscape, re-drawn               | the wadi corridor becomes a linear oasis — the geography the Romans' aqueducts and the old qsours all followed                                         | water in the oued = life along the banks = the south's original urban and agricultural map, restored                                                                                                                                                  |
| The upstream wound, healed by the sea | the headwaters of Gafsa's wadis rise across the border, where upstream dams stopped the historical flows — the oueds and the oases went dry from ABOVE | the loop's desal is the same order of magnitude as the lost flows (~55 Mm <sup>3</sup> /yr): a fraction released at the wadis' heads makes the south's rivers sovereign again — no upstream decision can turn them off, because the source is the sea |

**The gate:** the pilot oued: one release season measured end to end — infiltration rate, seed-bank response, downwind humidity, aquifer level before and after — the rain gauges and the piezometers decide the calendar, published like everything else. Release volume starts at the minimum the ecology answers to, and rises only with the measurements.

the answer to the three questions: possibility — yes, the dams and the oueds already exist, paid for; need — yes, three of them: the aquifer recharge (8-14 Mm<sup>3</sup>/yr of silent drought insurance), the ecosystem restoration (the pulse the banks evolved for), and the atmosphere (~3 Mm<sup>3</sup>/yr of vapor that the recycling band turns into clouds). The genius of it: ~20 Mm<sup>3</sup>/yr — a fraction of the loop's output — released on a calendar becomes the south's returning heartbeat: the oueds flow, the banks bloom, the aquifer rises, and the clouds come back. The dams stop being walls and become the pulse of the landscape they were built to hold. And the deep answer to the upstream wound: the wadis that died when the headwater dams closed them are reborn from the sea — the terraformation is not a copy of the old rivers, it is the country's first SOVEREIGN rivers, sourced where no upstream dam can reach.

## The batteries — gravity only, the land holding the night

| TIER                | THE INSTRUMENT                                                      | THE NOTE                                                        |
|---------------------|---------------------------------------------------------------------|-----------------------------------------------------------------|
| Tier 1 — seconds to | the pipeline's pressure — hydro accumulators on the artery, and the | no chemistry, no new civil works: the pipe itself is the spring |

| TIER                           | THE INSTRUMENT                                                                                                                                                           | THE NOTE                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| minutes                        | dispatch leg's 20 GWh/yr already booked                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                    |
| Tier 2 — the daily sun         | the PIPELINE BATTERY: pump-as-turbines on every descending stretch of the artery — the water transfer is the storage                                                     | the pumping energy is already in the water stack; the turbines recover it on the downhill runs — gravity gives back what the pumps paid                                                                                                                                                                                                                                            |
| Tier 3 — the long night        | the BARRAGE PAIR: two existing dams at 250 m class of head — 0.68 Mm <sup>3</sup> covers the whole night; one Mm <sup>3</sup> at this head stores ~531 MWh               | the civil works already exist and are paid for; only the penstock + pump-turbine are new (~\$36M at 30 MW) — 50-year life, zero degradation, zero supply chain. Phase 0's inventory names the pair: the Medjerda cascade (Sidi Salem 550 Mm <sup>3</sup> class), Sidi el Barrak (264 Mm <sup>3</sup> ) to the coastal plain, the Zaghouan line — the Roman aqueduct's own corridor |
| Tier 4 — the strategic reserve | FILL THEM ALL: ~1800 Mm <sup>3</sup> of the country's dam capacity stands empty after the drought years — the loop's desal + reuse refills it at ~55 Mm <sup>3</sup> /yr | a ~33-year program that restores the nation's water bank AND parks ~956 GWh of gravity energy in the country's own mountains — the dams were built to hold a country's water; now they hold its nights and its years too                                                                                                                                                           |

**The gate:** Phase 0's barrage inventory, dam by dam: head between neighbouring reservoirs, usable volume, fill state, silt, distance to the corridor — measured and published by the same rule. The pair with the measured head scales; the pond enters ONLY where the topographic survey says a pond-pair works — booked at zero until then.

the fill-them-all theorem: the loop's night is ~360 MWh — 0.68 Mm<sup>3</sup> at 250 m — while the country's dams hold ~1800 Mm<sup>3</sup> of empty, already-paid-for volume: a rounding error against the need, and a ~956 GWh gravity reserve waiting in the mountains. One Mm<sup>3</sup> at 250 m is ~531 MWh. The same water that ends the thirst carries the night: lift by day, flow by night, and let the empty barrages hold the country's future the way they were built to. Gravity is the only battery that never degrades, never burns, and was poured in concrete generations ago.

## The people's energy is electrons, not hydrogen — and the H<sub>2</sub> lives inside the loop

The loop's H<sub>2</sub> (2.63 kt/yr) is an industrial molecule: per kWh of heat, green H<sub>2</sub> at \$0.21 is 7× the loop's \$0.031/kWh electron; blending caps at 5-20% by volume (Pg20/Pg21/Pg22); the whole H<sub>2</sub> volume equals 1.8% of grid-gas demand. The molecule goes where only a molecule works: ammonia for the GCT chain, inside the loop. the electrolyzer sits on the desal line, and everything it touches stays in the loop: the water it splits is the SWRO permeate (0.04% of the 150,000 m<sup>3</sup>/d output); its un-split reject stream is clean water that returns to the product line; its oxygen by-product aerates the spirulina ponds and greenhouses; its stack heat (15–20% of input) joins the kiln and DC reject heat on the desal preheat circuit; and the minerals concentrated in its flush stream join the brine valorization line. The electrolyzer is not a consumer of the loop — it is another recovery point inside it.

## The total routing — every stream gets a home

The ponds do *not* drink brine: SWRO brine is 55-70 g/L, past *Arthrospira*'s ~30 g/L ceiling (Pg31); they drink the desal permeate and breathe the electrolyzer's oxygen. The full inventory:

| STREAM                                   | PRODUCED BY                | ITS HOME                                                                                             | THE NUMBER                                   |
|------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Solar electrons                          | 500 MW fleet               | desal, metals, DCs, electrolyzer, pumps — routed by the dispatcher                                   | 876 GWh/yr                                   |
| SWRO permeate                            | coastal desal              | city offtake · electrolyzer feed · ponds · greenhouses                                               | 150,000 m <sup>3</sup> /d                    |
| SWRO brine                               | coastal desal              | salt + magnesium valorization, evaporated on the loop's own free heat                                | 122,727 m <sup>3</sup> /d → 58 kt Mg/yr      |
| Brine bittern                            | valorization               | Mg(OH) <sub>2</sub> fire retardant → <b>the board's own chemistry</b>                                | 140 kt/yr = 140% of the board's need         |
| Electrolyzer oxygen                      | H <sub>2</sub> plant       | spirulina pond + greenhouse aeration                                                                 | 21 kt/yr                                     |
| Electrolyzer hydrogen                    | H <sub>2</sub> plant       | ammonia → the GCT fertilizer chain                                                                   | 2.63 kt/yr                                   |
| Electrolyzer reject water / heat / flush | H <sub>2</sub> plant       | product line / desal preheat / brine line                                                            | nothing leaves                               |
| DC reject heat                           | compute park               | the same seawater pipe, arriving warm at the membranes                                               | −\$0.007/m <sup>3</sup>                      |
| Kiln + acid exotherm                     | board kilns + metals plant | desal preheat + greenhouses                                                                          | 30 GWh/yr                                    |
| CO <sub>2</sub> (kiln + power)           | combustion streams         | mineralized into board (2 Mt PG) · pond carbon                                                       | 540 t CO <sub>2</sub> /yr to the ponds       |
| Phosphogypsum                            | GCT stack                  | fire-rated board — with its retardant from our own brine                                             | 2 Mt/yr                                      |
| Acid stream                              | GCT wet-process acid       | U + heavy REE extraction; raffinate back to fertilizer                                               | 716.5k lbs U <sub>3</sub> O <sub>8</sub> /yr |
| Municipal return flows                   | city mesh                  | 30%+ reuse · pump-as-turbines on the descent                                                         | 3.9 GWh/yr                                   |
| Sludge / digestate                       | wastewater + organics      | biogas for kilns and greenhouses                                                                     | booked option, honestly labeled              |
| Slaughterhouse bones                     | local slaughterhouses      | bone char → the fluoride filters; <b>spent char</b> → <b>phosphate soil amendment for the fields</b> | filter loop closes into fertilizer           |
| Spent pond medium                        | spirulina harvest          | nutrient-rich → greenhouse fertigation                                                               | nothing discarded                            |
| Ammonia fertilizer                       | GCT chain                  | the terfas fields' host plants + greenhouse soils                                                    | the truffle grows on the loop's own nitrogen |
| Compute services                         | colo park                  | export — the data product                                                                            | tenant-gated                                 |
| Ra-226 / Po-210                          | acid stream radiology      | stays in the gypsum/board matrix — the radiology gate                                                | below the 1,000 Bq/kg exemption              |
| Board offcuts                            | board line                 | recycled into new board                                                                              | zero waste                                   |
| Sodium chloride                          | brine line                 | industrial offtake at the park gate                                                                  | nothing hauled                               |



| STREAM                        | PRODUCED BY | ITS HOME                                               | THE NUMBER                            |
|-------------------------------|-------------|--------------------------------------------------------|---------------------------------------|
| Halophytes / Artemia on brine | brine line  | future option, honestly labeled — not a current credit | the only stream awaiting its consumer |

the ponds drink the desal permeate —  $\sim 0.2\%$  of the  $150,000 \text{ m}^3/\text{d}$  output — and breathe the electrolyzer's oxygen; the brine (55–70 g/L, far past the  $\sim 30 \text{ g/L}$  ceiling the organism tolerates) is routed to salt and magnesium valorization instead. Halophytes and Artemia on brine are booked as a future option, honestly labeled — not a current credit.

## The fuel invoice, honestly split — the 500 MW is the grid's medicine, not the whole prescription

of the \$4.895B fuel invoice, the 500 MW fleet displaces only the power-sector gas slice —  $\sim \$0.75\text{B}/\text{yr}$  (DERIVED from the energy balance) — by  $\$106\text{--}125\text{M}/\text{yr}$ , which is 14.1–16.7% of that slice. The rest of the invoice has its own answers, not solar's: the bottles ( $\sim \$0.55\text{B}$  of LPG, the bonbonne) leave the market as the people shift to electrons — induction on the loop's own surplus — and the biogas pilot; transport fuels ( $\sim \$2.6\text{B}$ ) are the honest frontier. The 500 MW is the grid's medicine, not the country's entire prescription.

## The mountains, honestly analyzed

Fog collection is real where fog lives (10–20 L/m<sup>2</sup>/day, Pg33) — Tunisia's fog lives in the already-humid NW; the truffle mountains are the arid interior, and cloud seeding has no proven routine gain (Pg32). What the mountains give the loop: the mountains enter the loop as storage, rainfall and free cooling — and as the land where the truffle belongs (freeing the lowlands that grow today's food). Fog harvesting from pipe meshes stays off the program: the moisture is not there to collect, and the honest scientist does not plant a net under a cloudless sky.

## The AI regulation — five layers, down to the drippers

The dispatcher is a model-predictive controller on a 15-minute receding horizon over the mass balances (Lexicographic: safety > contracts > economics); the twin trains it offline and audits it online, with a Kalman filter for unmeasured states and a published prediction-error KPI; node controllers stay PID-class with hard-wired interlocks; the governance layer never delegates a gate.

| LAYER                                                    | WHAT IT DOES                                                                                                                                                                                                                                                       | THE MATHEMATICIAN'S NOTE                                                                                                   |
|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Layer 0 — the instrumented field                         | every node is a sensor + an actuator: membrane pressure and conductivity, brine concentration, pond pH and temperature, electrolyzer stack voltage, kiln thermocouples, PAT speed, reservoir levels, household-filter telemetry, the grid tie, the weather station | the stream inventory above is the measured state — the mass balances are the constraints the controller must never violate |
| Layer 1 — node controllers (edge, millisecond-to-second) | local closed loops of PID class: membrane pressure, pond dosing, kiln temperature, pump speed — with hard-wired safety interlocks                                                                                                                                  | functional safety is never on the AI: a controller that fails safe is a relay, not a model                                 |

| LAYER                                                                           | WHAT IT DOES                                                                                                                                                                                                                                              | THE MATHEMATICIAN'S NOTE                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Layer 2 — the dispatcher (model predictive control, 15-minute receding horizon) | a multi-variable MPC: state = solar forecast, reservoir inventory, brine concentration, electrolyzer turndown, DC load, grid price, offtake commitments; decisions = pump flows, setpoints, brine routing, dispatch; forecasts 36–48 h weather, 24 h load | objective: minimize loop energy cost + degradation + gate-violation penalties, subject to mass balances, storage states and contract floors — lexicographic, never weighted: safety > contracts > economics. Its measured output is the 20 GWh/yr dispatch value and the routing-dependent water credits |
| Layer 3 — the digital twin (simulation, learning, audit)                        | a physics-calibrated model of every balance: trains the dispatcher offline, audits every decision after the fact, simulates before any physical change                                                                                                    | unmeasured states (membrane fouling, pond bloom state) are estimated with a Kalman-class filter from pressure-flow and optical residuals; the twin's prediction error is a published KPI — the AI's accuracy is measured, not claimed                                                                    |
| Layer 4 — the governance layer (the boundary, unchanged)                        | the scoreboard computes and publishes every gate number from Layer 0 telemetry with provenance; the price index; the results framework; anomaly flags to humans                                                                                           | the AI never sets a gate, never signs, never prices — the falsification ladder is mechanical thresholds, not a learned model. The nervous system is a servant, not a master                                                                                                                              |

| NODE            | SENSES                                                            | CONTROLS                                                                        | TIMESCALE          | LAYER |
|-----------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------|-------|
| PV field        | irradiance, string current, panel temperature, soiling sensors    | inverter setpoints, curtailment, cleaning scheduling                            | seconds–minutes    | L1–L2 |
| Desal trains    | feed pressure, permeate conductivity, ERD pressure, CIP condition | high-pressure pump, ERD bypass, antiscalant dosing, CIP cycles                  | seconds–hours      | L1–L2 |
| Membrane health | pressure-flow residuals (fouling inferred by Kalman filter)       | maintenance scheduling, train rotation                                          | days               | L3    |
| Brine line      | concentration, temperature, Mg saturation                         | evaporator feed rate, Mg(OH) <sub>2</sub> precipitation setpoints               | minutes            | L1–L2 |
| Electrolyzer    | stack voltage, current, purity, water feed                        | turndown, ramp rate, O <sub>2</sub> vent routing to ponds, reject routing       | seconds            | L1–L2 |
| Board kilns     | temperature profile, feed rate, offcut fraction                   | burner setpoints, retardant dosing, offcut recycle                              | minutes            | L1–L2 |
| Metals plant    | SX stream assays, raffinate quality                               | extraction/stripping setpoints, raffinate return to fertilizer                  | minutes–hours      | L1–L2 |
| DC park         | cooling-loop temperature, PUE, workload mix                       | seawater flow to racks, batch-workload scheduling to solar peaks                | minutes            | L2    |
| Water mesh      | reservoir levels, pump status, pressure heads, PAT speed          | pump scheduling, zone valves, PAT bypass, the solar-following curve             | 15-minute dispatch | L2    |
| Spirulina ponds | pH, salinity, temperature, optical density                        | aeration timing, CO <sub>2</sub> injection, medium exchange, harvest scheduling | hours              | L1–L2 |

| NODE                                | SENSES                                                                            | CONTROLS                                                                                                                                                                          | TIMESCALE        | LAYER     |
|-------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------|
| Greenhouses                         | EC, pH, humidity, CO <sub>2</sub> , soil moisture                                 | fertigation dosing (spent medium), ventilation, CO <sub>2</sub> enrichment from the kiln stream                                                                                   | minutes          | L1–<br>L2 |
| Establishment drip — to the emitter | per-zone capacitance soil-moisture probes, per-zone flow meters, weather forecast | zone valve actuators per management block, pump pressure, fertigation dosing — emitters stay passive pressure-compensated; clogged drippers detected from per-zone flow residuals | hourly–<br>daily | L1–<br>L2 |
| Household filters                   | usage telemetry, replacement flags                                                | fleet logistics: replacement routing, spare inventory, the pilot's uptime KPI                                                                                                     | days             | L2–<br>L4 |
| Grid tie                            | import/export price, solar forecast, offtake commitments                          | import vs. dispatch decisions                                                                                                                                                     | 15-minute        | L2        |
| Scoreboard + cohort                 | every KPI stream above, provenance-hashed                                         | gate computation, price index, anomaly flags to humans                                                                                                                            | continuous       | L4        |

**what the AI is honestly worth: the 20 GWh/yr dispatch (~\$0.8M/yr) + the routing-dependent water credits it executes (~\$0.05-0.08/m<sup>3</sup> — DC heat, kiln heat, brine heat, dispatch) + a DERIVED 1–3% park OPEX from multi-variable coupling no manual schedule can follow — and the scoreboard, which is the program's entire credibility.** if the AI is down, the loop does not stop: Layer 1 runs every node in safe, suboptimal mode — the degraded cascade already computed (\$0.521/m<sup>3</sup> water, gate still clear). AI uptime is a KPI, not a dependency.

## The irrigation — to the dripper, honestly

the fields drink at establishment, then live on rain — and the greenhouse loop runs on the pond's own spent nutrients. The irrigation system is designed the way the truffle itself is: drought-proof by architecture, not by subsidy. Establishment: 624 m<sup>3</sup>/ha/yr for the first two years → 4,274 m<sup>3</sup>/d for 2,500 ha, from inland brackish water and reuse; then rainfed.

## Cloud generation — the honest version

our 12 km<sup>2</sup> array absorbs ~156 MW more sunlight than the bare soil (albedo Δ0.05) — a persistent local heat source, plus the ponds' evaporation placed upwind of the fields. Li et al. (Pg34) measured the class of effect at continental scale; at ours it is a designed microclimate, not a rain machine: soil shaded by panels retains moisture, the updraft adds marginal convective activity, and the dew that follows is free water on the host plants. The honest claim is directional and modest — and it costs nothing extra: the array was being built anyway.

## The mountain portfolio — if mountain agriculture is needed, the version that scales

| CROP / SYSTEM             | HECTARES       | WATER                            | VALUE                               | THE SCALE LOGIC                                   |
|---------------------------|----------------|----------------------------------|-------------------------------------|---------------------------------------------------|
| Terfas belt (arid jebels) | 2,500-3,000 ha | rainfed + micro-catchments + dew | €15-45M/yr at scale (market-capped) | already the design: the drought-proof luxury crop |

| CROP / SYSTEM                                          | HECTARES                                                 | WATER                                                                            | VALUE                                                                          | THE SCALE LOGIC                                                                                                     |
|--------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|                                                        |                                                          |                                                                                  |                                                                                | on land nothing else farms                                                                                          |
| Cactus pear (opuntia) belt                             | consolidate + expand the existing ~600k ha national belt | zero irrigation; 300-500 mm survives on 150                                      | fruit at 20-40 t/ha for the basin's price index + cladodes as livestock fodder | the arid mountain staple — Tunisia is already a world producer; the program adds value chains, not water            |
| Medicinal & aromatic plants (rosemary, thyme, oregano) | 5,000-10,000 ha across the Dorsale                       | rainfed; 300-500 kg/ha dry                                                       | ~€8-20M/yr export at €4/kg dry                                                 | the highest value-per-kilo crop that needs no water at all — the mountain cash crop #2                              |
| Agrivoltaic fodder (sulla/alfalfa under the PV field)  | up to 1,250 ha of shaded soil under the 500 MW array     | the panels' shade halves evaporation; no added water beyond the field's own rain | fodder for ~5,000-10,000 head; the flock fertilizes the field                  | the loop's livestock leg: sheep graze solar parks already (Spain/US precedent) — the AI routes the grazing schedule |
| Rainwater harvesting (swales + terracing)              | 2-3% of mountain catchment area                          | each ha of swales captures ~3,000 m <sup>3</sup> /yr at 300 mm rainfall          | \$200-500/ha — an order of magnitude under drip-system capex                   | the mountains' real water system: catch the rain that already falls there, at scale, without pipes                  |

mountain agriculture at scale means putting each crop where its water cost is zero: terfas, cactus, MAPs and agrivoltaic fodder on the mountains' own rain and the loop's own shade — while wheat, vegetables and orchards stay on the lowlands and in the greenhouses where water is abundant. The mountains feed the people what only mountains can grow; the pipes carry water only where gravity and economics both say yes.

## The growroom at landscape scale — climate steering toward bio-recovery

**Measures:** Atmospheric transect: RH / temperature / dew-point stations up the slope at the mist corridor, leaf-wetness sensors, 2-3 sonic anemometers for the anabatic flow; Soil grid: capacitance moisture probes per management zone (the drip grid doubled as the climate grid), soil temperature; The sky: satellite NDVI every 5 days into the twin, rain gauges along the corridor, pond evaporation pans; The park: panel albedo/soiling sensors (the field's reflectivity IS a climate actuator), pond water temperature

**Steers:** The mist corridor: fog nozzles, anabatic hours only, virtual-temperature-positive dosing — 2% of desal flow; The shade: the PV array's own footprint: shaded soil retains moisture; cleaning schedule shifts albedo; The pond surface: evaporation timing follows the wind — the ponds breathe upwind of the fields on purpose; The swales: rainfall harvested at contour, released by zone valves into the host-plant belt

**Objective:** the bio-recovery term in the dispatcher's objective: maximize dew days + soil-moisture persistence above wilting point + NDVI slope — subject to the energy budget, the thermals remaining buoyant, and the rain-gauge referee. The twin adds a soil-water-balance module (Penman-Monteith evapotranspiration, standard equations, computed in code) so every steering decision has a predicted and a measured outcome.

**The spiral:** the feedback the steering rides: mist → dawn dew → soil moisture → host plants establish → transpiration raises local humidity → more dew — the humidity loop that turns a slope back green, one

feedback cycle at a time. The KPIs published: dew days per month, days above wilting point, NDVI slope — bio-recovery, measured like a growroom, audited like everything else.

## The invisible drop — and the effects ledger, direct and side

the drop → microbial pulse → nutrient mineralization → plant-available N and P → host-plant growth → nectar and pollen → the insects: wild bees at 1,000-2,000 flower visits a day, ants engineering the soil, beetles on the wind → the host plants set seed (Helianthemum is insect-pollinated) → the mycorrhizal trade persists → the truffle → the harvest → the price index. Every dew morning the mist buys is  $10^8$  cells per square centimeter of leaf waking up — and the corridor's vegetation is the pollinator highway the truffle's own host plants depend on. One invisible drop feeds the microbes that feed the plants that feed the insects that pollinate the plants that feed the truffle that feeds the people.

and the drop re-enlists the army Tunisia lost: the right-of-way vegetation is the refuge for the beneficials — ladybirds (5,000 aphids each in a lifetime), lacewings, hoverfly larvae (400-600 aphids each), and the parasitoid wasps Tunisia already proved at national scale in the date oases (Trichogramma against the date moth). Every predator the corridor hosts is a pesticide that was not bought, not sprayed, and not inhaled by a child — biocontrol is the health ledger's quietest leg.

| EFFECT                                  | CLASS                | THE MODEL                                                           | THE INSTRUMENT                       |
|-----------------------------------------|----------------------|---------------------------------------------------------------------|--------------------------------------|
| Mist vapor transport                    | DIRECT               | Gaussian plume + anabatic advection                                 | sonic anemometers, RH transect       |
| Dew deposition on host plants           | DIRECT               | surface energy-balance dew model ( $T_{2surf} < \text{dew point}$ ) | leaf-wetness sensors, dew gauges     |
| Soil moisture gain, corridor zone       | DIRECT               | Penman-Monteith water balance                                       | capacitance probe grid               |
| Seedling establishment survival         | DIRECT               | water-stress survival curves                                        | nursery registry, field counts       |
| PV-shade moisture retention             | DIRECT               | shade fraction × PET reduction                                      | soil probes under panels             |
| Thermal buoyancy (the constraint)       | DIRECT               | virtual-temperature profile                                         | T/RH + wind stations on the slope    |
| Microbial Birch pulse                   | SIDE                 | rewetting respiration curves                                        | soil CO <sub>2</sub> efflux chambers |
| Beneficial insect recruitment           | SIDE                 | habitat-area × predator-prey dynamics                               | sticky traps, acoustic/AI counters   |
| Pollinator visitation → host seed set   | SIDE                 | pollination-saturation curves                                       | visit counts, seed counts            |
| Precipitation return                    | SIDE — ZERO-credited | cloud microphysics (the least-solved field, Pg32)                   | rain-gauge network                   |
| Pesticide displacement → child exposure | SIDE                 | exposure-response from the cohort                                   | urinary pesticide metabolites        |
| Atmospheric dust & PM scavenging        | SIDE                 | wet-deposition scavenging coefficients (dew/fog wash)               | PM sensors along the corridor        |

| EFFECT                  | CLASS  | THE MODEL                              | THE INSTRUMENT                                     |
|-------------------------|--------|----------------------------------------|----------------------------------------------------|
| Air quality, basin-wide | DIRECT | emissions inventory → dispersion model | PM2.5/NOx/SO2 grid + the cohort's respiratory KPIs |

## The land, projected — the Darwinian view of the environment itself

| LAYER OF LIFE    | WITHOUT THE PLAN                                                                                                                  | WITH THE PLAN                                                                                                                                      |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Soil biology     | the seed bank dies with the soil — desertification's positive feedback: less vegetation, less humidity, less dew, less vegetation | the Birch pulse wakes with every dew morning; the microbial city rebuilds; 10 km of hyphae per gram of soil come back to trade                     |
| The insects      | beneficials stay locally extinct; the pesticide treadmill accelerates; pollinators thin, host plants set less seed                | the right-of-way is the refuge: ladybirds, lacewings, hoverflies, Trichogramma's cousins re-recruit; every predator is a pesticide not bought      |
| The dew economy  | bare, crusted ground sheds what little rain falls; the drop dies on arrival                                                       | the mist corridor turns the dawn into a harvest: dew on host plants, soil moisture banked, the spiral compounding                                  |
| The truffle belt | the wild harvest shrinks; the mycorrhizal network starves                                                                         | the belt establishes on the corridor's moisture: 100 ha, then 2,500-3,000 ha, on the land the humidity itself prepared                             |
| The frontier     | desertification marches downhill — the slope's fitness landscape selects for stone                                                | each humidified season pushes the establishment frontier a little further upslope — the landscape's selection flips from bare ground to vegetation |

the Darwinian projection of the whole machine: the same land, two futures — one where the environment keeps selecting for stone, and one where a 2% mist margin re-engineers the fitness landscape so that life out-competes bare ground. The honest bound stands: the corridor transforms a belt, not a mountain range — the wider slopes change only through the slow, compounding spiral and the rain cycle. We plant the drop; the land grows the rest.

## The nature ledger — every class of credit, not just carbon

| SPECIES / ASSET                        | THE HONEST NOTE                                                                                                                 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Wild bee community                     | ~700+ species documented in Tunisia; the corridor's nectar belt is their refuge and the truffle host plants' pollination engine |
| Egyptian vulture (globally endangered) | breeds in Tunisia; scavenger of the open steppes the corridor stabilizes                                                        |
| Dorcas gazelle                         | endangered; the restored steppe of the south is its historic range                                                              |
| Cork oak + orchid flora (NW)           | the humid mountains' own endangered flora — the value-chain program that ends over-harvesting                                   |
| Medicinal & aromatic plants            | wild-rosemary pressure relieved by the cultivated MAP belt — the market saves the wild stock                                    |

| SPECIES / ASSET | THE HONEST NOTE                                                                                      |
|-----------------|------------------------------------------------------------------------------------------------------|
| Addax           | honestly named: locally extinct in the wild — NOT claimed; restoration is a later, separate question |

the nature ledger: the corridor's ~50 km<sup>2</sup> of restored belt at maturity earns **\$0.0-0.2M/yr** of biodiversity credits at the nascent-market band (\$5-50/ha, DERIVED — booked at the low end; TNFD disclosure framework, Pg37; GBF Target 19, Pg36) and unlocks a **debt-for-nature swap**: \$500M of the state's debt refinanced against measured nature outcomes — ~\$7.5M/yr of coupon savings, the Gabon/Ecuador instrument (Pg38). The MRV system already exists: the acoustic counters, the NDVI feed, the soil grid — the same instruments that steer the mist also certify the credits. Not counted in the FX bridge (nature credits are not FX-grade exports): this is the program's second ledger, and the scoreboard publishes it the same way.

## The credit spectrum — every financial credit the country earns

| CLASS                       | WHAT IT IS                                      | THE HONEST NUMBER                                                                                    | STATUS      |
|-----------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------|
| The gas-displacement credit | the 500 MW solar farm                           | \$106-125M/yr of the power-sector gas slice (14-17% of the ~\$0.75B slice, not the \$4.895B invoice) | STRONG      |
| The dispatch value          | the MPC dispatcher routing every joule          | 20 GWh/yr of load shift + 33.9 GWh/yr recovered                                                      | STRONG      |
| The gravity return          | pump-as-turbines on the artery's descents       | 3.9 GWh/yr at 70% of flow returning at 50 m head                                                     | DERIVED     |
| The brine-pressure credit   | the isobaric ERD inside the membranes           | kept inside the 4 kWh/m <sup>3</sup> SWRO benchmark, carried as zero                                 | ZERO        |
| The heat credit             | the datacenter's waste heat preheating the feed | ~\$0.007/m <sup>3</sup> of energy saved — and biofouling accelerates, stated                         | DERIVED     |
| The oxygen credit           | the electrolyzer's O <sub>2</sub> to the ponds  | aeration cost avoided at the spirulina ladder                                                        | TIERED      |
| The carbon credit           | the kiln's CO <sub>2</sub> to the pond carbon   | ~540 t/yr of pond carbon feed                                                                        | DERIVED     |
| The magnesium credit        | the brine's Mg to the board's fire retardant    | ~58 kt/yr Mg vs the board's need — the loop's best closure                                           | STRONG      |
| The biodiversity credit     | the corridor's certified fauna                  | \$0.025-0.25M/yr, low band on purpose                                                                | SPECULATIVE |
| The water-reuse credit      | the reuse leg serving the oases and the fields  | 30%+ reuse turns the brackish band into the \$0.34/m <sup>3</sup> class                              | STRONG      |
| The aquifer credit          | the oued pulse's infiltration                   | 8-14 Mm <sup>3</sup> /yr of silent recharge — the KPI's rise                                         | TIERED      |
| The peace dividend          | the loop's share of the phosphate recovery      | phosphate 8 → 3.1 Mt = ~\$0.5-0.7B/yr of lost export value — the state's killer number               | STRONG      |

the spectrum, graded: the credits are the margin, not the plan — STRONG where the instrument exists, DERIVED where the arithmetic stands, SPECULATIVE or ZERO where honesty demands it.

## The economic-closure test — the physical flows close; the economic flows close by contract

| STREAM                | BUYER OF RECORD                                                   | THE CONDITION                                                                              |
|-----------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Water                 | SONEDE offtake at the gate price                                  | the \$0.68 gate clears even in the degraded cascade — the buyer is the state's own utility |
| Solar                 | STEG PPA at the tendered \$31/MWh class                           | the Tunisian tender precedent is the buyer's credit quality                                |
| Uranium               | a Cameco-class buyer under IAEA safeguards                        | ZERO until the pilot's assay + signed off-take                                             |
| REE concentrate       | a separation-capable offtaker                                     | ZERO until the separation-gap economics are priced                                         |
| Mg(OH) <sub>2</sub>   | the board (internal) + the surplus at the park gate               | internal offtake at avoided cost; surplus ZERO until signed                                |
| Terfas                | the domestic quota at EUR12/kg (contractual) + export at EUR20-40 | gated on 3 consecutive seasons ≥200 kg/ha                                                  |
| Spirulina             | the canteens + the feed-mill offtake                              | the food is proven; the volume is gated on signed offtake                                  |
| Biodiversity & carbon | a named credit buyer under TNFD/GBF frameworks                    | booked at the LOW band; MRV = the corridor's own sensors                                   |

  

| POINT                         | WHAT IT DEMANDS                                                                               | WHERE IT LIVES     |
|-------------------------------|-----------------------------------------------------------------------------------------------|--------------------|
| Concentration, measured       | the acid stream's actual U/REE assay — the IAEA/Gabès anchors bracket it; Phase 0 measures it | Phase-0 assay #1   |
| Recovery rate, at scale       | DEHPA/TOPO is the proven chemistry (GCT study); the pilot's yield is the gate                 | Phase-0 assay #2   |
| Reagent consumption           | solvent losses per tonne — the dominant opex; benchmarked against the published SX economics  | Phase-0 assay #3   |
| Separation cost, REEs         | concentrate ≠ separated oxide — the page's own register states it; the pilot prices the gap   | Phase-0 assay #4   |
| Rad-waste pathway             | Ra stays in the gypsum/board matrix; the radiology gate audits it                             | gate, audited      |
| Product specification         | yellowcake grade + separated-oxide spec — named, not assumed                                  | gate, contractual  |
| Buyer of record               | a signed off-take, not a market price — the leg's revenue books at ZERO until it exists       | gate, signed       |
| Transport & export compliance | IAEA safeguards chain + the CNSTN/Cameco-class controls                                       | gate, compliance   |
| Permitting                    | the mining/waste-reprocessing license stack — named in the gates                              | gate, legal        |
| Capex & opex, EPC-grade       | the pilot's cost curve becomes the EPC estimate — nothing scales before it                    | gate, financial    |
| Price sensitivity             | the leg's grade bands (STRONG/PLAUSIBLE/SPECULATIVE) restate at every price shock — published | engine, continuous |



| POINT              | WHAT IT DEMANDS                                                                                                               | WHERE IT LIVES          |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Independent review | the whole chain is the invitation: attack the assay, the yield, the separation gap — the falsification ladder is built for it | hostile review, invited |

the economic-closure test: the physical flows close by construction — the economic flows close by CONTRACT, one named buyer at a time. Every revenue stream above has exactly one condition that moves it from model to money, and the page books each stream at zero until that condition exists. That is the answer to the central question: the closure is not assumed, it is the falsification ladder itself — the 12-point metals chain is the specimen, and every other leg wears the same harness.

## The capital-formation pack — dual-track financing, banks first, blockchain as the expansion rail

| STAGE                              | THE NUMBERS                                                                             | THE COVERAGE                                             |
|------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------|
| Year 1 — Phase 0 + pilots          | revenue near zero; the \$1.5M Phase 0 + the \$3-6M pilots run on grant/equity           | DSCR n/a — de-risking capital, not debt                  |
| Year 2-3 — solar + water on stream | solar PPA \$27.2M + water margin \$10.9M → EBITDA ~\$38M                                | DSCR 1.16× on the utility legs alone                     |
| Year 4-5 — metals pilot gated in   | +\$35M mid-grade metals → EBITDA ~\$73M                                                 | DSCR 2.22× — the floor cleared, the token tranche serves |
| Year 5+ — full loop                | brine valorization, dispatch, terfas and nature-ledger cash join in measured increments | each gate passed re-prices the junior in public          |

| RUNG                   | WHAT IT COVERS                                                                  |
|------------------------|---------------------------------------------------------------------------------|
| 1. Offtake revenues    | SONEDE/STEG offtake + solar PPA — state-backed, the senior's cover              |
| 2. O&M                 | operating costs before any capital return — the operators' lien                 |
| 3. Senior debt service | traditional banks / IFC-WB-EBRD syndicate — DSCR floor 1.25×                    |
| 4. Reserve             | 6 months of service — the trustee's cushion, enforced in law                    |
| 5. Token coupons       | the junior/revenue-participation tranche — KPI-linked via the scoreboard oracle |
| 6. Junior & equity     | the risk capital — first in, last out                                           |

| INSTRUMENT        | THE SPECIFICATION                                                                                                                                          |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The SPV           | offshore (ADGM, the regional standard) — bankruptcy-remote issuer, Tunisia has no crypto framework; the project stays Tunisian, the token's issuer doesn't |
| What is tokenized | revenue-participation rights on the program's utility-grade legs (water + solar), NOT the park, NOT the land, NOT the state's sovereignty                  |
| Brickken gate     | KYB/KYC, legal opinions, valuation, cap table — the issuance platform's DD gate, item by item                                                              |
| Centrifuge pool   | the SPV's RWA pool with monthly NAV computed from the scoreboard — senior/junior tranches, trustee-controlled                                              |

| INSTRUMENT               | THE SPECIFICATION                                                                                                                                       |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Morpho Blue rail         | the PPA receivables as collateral: SONEDE/STEG offtake quality → LLTV ~60-70%, oracle = the scoreboard's signed revenue feed                            |
| The oracle               | the page's own public KPIs (water quality gates, health cohort, price index) → signed feed → on-chain — the falsification gates become smart conditions |
| Compliance               | Reg S for offshore, MiCA classification for EU, Travel Rule, Tunisian capital-account mechanics via the BCT — named, not assumed                        |
| The state's contribution | offtake + land + feedstock, valued and credited — the state puts in what only it can, and the waterfall repays it in the same currency                  |

| PARTY                     | THEIR ROLE                                                                                                          |
|---------------------------|---------------------------------------------------------------------------------------------------------------------|
| The State                 | of-take counterparty + land + feedstock + the permits — the credit foundation everything else prices off            |
| The Investors             | senior lenders (WB/IFC class) + the token holders — two rails, one project                                          |
| The Engineers & Operators | the EPC + O&M — their lien sits at waterfall rung 2                                                                 |
| The Health & Community    | the cohort, the cooperatives, the canteens — the KPIs the token's coupons are LINKED to                             |
| The SPV (ADGM)            | the issuer — bankruptcy-remote, the token's legal home                                                              |
| The Trustee               | the reserve and the waterfall, enforced in law not code — the code can fail, the trust deed cannot                  |
| The Oracle                | the scoreboard bridge — the sensors we already built are the only oracle we need                                    |
| The Platform              | Brickken for issuance, Centrifuge for the pool, Morpho for the collateral rail                                      |
| The Auditors              | code audit + financial audit + KPI verification — the trifecta this critique era taught us                          |
| The land & the fauna      | the nature ledger's beneficiary — the KPI-linked structure turns the restored corridor into a tokenized public good |

**The honest capital math:** the honest capital math: the utility-grade legs alone produce ~\$38M/yr of EBITDA against ~\$33M/yr of service on \$300M senior — DSCR 1.16×, BELOW the house's 1.25× floor; with the metals pilot at its mid grade it becomes 2.22×. Valuation anchor: \$228-304M of equity value at 6-8× the utility-grade EBITDA at maturity — the token tranche's exit multiple, stated before any token exists. The conclusion is the discipline itself: senior debt covers the bankable legs, the token tranche funds the DERISKING — every Phase-0 gate passed raises the DSCR in public, and the token's coupons rise with the measured KPIs. Banks first; the blockchain is the expansion rail, not the foundation.

## The MIGA wrap — borrowing above the country's rating

| COVERAGE             | WHAT IT MEANS FOR THE PROGRAM                                                      |
|----------------------|------------------------------------------------------------------------------------|
| Transfer restriction | the dinar cannot leave — MIGA compensates; the FX bridge's worst case is insured   |
| Asset-seizure risk   | the state takes the assets — MIGA compensates; the investor's deepest fear, priced |

| COVERAGE                 | WHAT IT MEANS FOR THE PROGRAM                                                                         |
|--------------------------|-------------------------------------------------------------------------------------------------------|
| Instability events       | the region's oldest risk — covered, so the peace dividend's opposite has a floor                      |
| Contract non-performance | the state breaks the offtake/PPA — covered; the water and solar buyers become bankable counterparties |

**The mathematics of the wrap:** the MIGA wrap, computed: the program's \$300M senior tranche, guaranteed, prices at AAA — service falls from \$32.9M/yr (B- class, 7%) to \$27.9-28.9M/yr (4.5-5%, the MIGA-guaranteed band, Pg49) — and the base-case DSCR rises from 1.15× to 1.36-1.31×: the utility-grade legs ALONE clear the 1.25× floor. That is the single most important number MIGA changes: the project becomes bankable on water and solar, without waiting for the metals — the metals then de-risk from strength, not from necessity. The wrap does not raise Tunisia's rating; it lets the project BORROW above it — and that is exactly what the capital stack needed.

## The datacenter's own capital path — other people's credit does the heavy lifting

| RUNG                                | THE NUMBERS                                                                                                                                                       | THE FINANCING                                                                                      |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| The pilot: 11.3 MW colocation       | capex ~\$40M — the page's own number                                                                                                                              | financed by anchor tenants + vendor equipment leases, NOT the program's debt                       |
| The scale: 100 MW, hyperscale-class | capex \$3.0-4B of GPUs — the review's own register                                                                                                                | a SEPARATE tenant-backed SPV: the anchor leases carry the GPUs; the program never borrows for GPUs |
| The instruments                     | vendor deferred payment (GPU-financing class) + anchor-tenant prepay + colocation build-out debt against signed leases + export-credit agencies for the equipment | each instrument names its own collateral — none of it is the water or the health program           |
| The loop's role                     | the loop SELLS the DC its advantages: \$0.031/kWh power, the water mesh, seawater cooling (PUE 1.1-1.2 class) — the DC's margin is the loop's offtake             | the solar PPA revenue already counted in the stack IS this leg                                     |

| THEOREM             | THE COMPUTATION                                                                                                                                                                                                             | THE NOTE                                           |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| The TCO arbitrage   | a 100 MW DC at EU-grid power and air cooling burns 1,270 GWh/yr = \$140M; on the loop it burns 1,007 GWh = \$31M — the co-location surplus is <b>\$108M/yr</b>                                                              | measured from the loop's own PPA and PUE anchors   |
| The surplus split   | the loop and the SPV bargain over \$108M/yr; the Nash solution splits it at the disagreement midpoint — the loop's share ~\$54M/yr above the base PPA, the DC keeps the rest — both strictly better off than building apart | the bargaining range is the loop's pricing power   |
| The deflation hedge | AI compute prices fall ~20-30%/yr — the payback window closes fast; anchor-tenant FIXED-price leases are the hedge, and they are the SPV's risk, not the program's                                                          | the program carries only the utility legs it sells |

the DC's honest capital path, in the Nobel register: at \$2000M/yr of revenue class per 100 MW against \$3.0B of GPUs, the payback is ~1.5 years IF utilization and prices hold — which is precisely why the program does not carry it. What the loop carries is the ARBITRAGE it creates: \$108M/yr of genuine co-

location surplus, split by negotiation, captured at the PPA and the cooling fees. The compute leg's FX contribution stays in the speculative tier until an anchor lease exists; the DC's financing is the one place where other people's credit does the heavy lifting — and the loop's position is the only one in the stack that gets stronger with every megawatt someone else pays for.

## The strip-down — the critique's killer question, answered

| MODULE                  | THE STAND-ALONE ECONOMICS                                                                                                     | THE VERDICT               |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Water                   | stand-alone at the degraded \$0.521/m <sup>3</sup> cascade — under the \$0.68 SONEDE gate even with every soft credit at zero | STANDS ALONE              |
| Solar                   | stand-alone at the tendered \$31/MWh PPA class — no coupling required                                                         | STANDS ALONE              |
| Uranium                 | stand-alone IF the Phase-0 assay holds — the extraction is industrial precedent, the Tunisian stream is the measurement       | STANDS ALONE, GATED       |
| Terfas                  | stand-alone at the 3-season ≥200 kg/ha gate — zero-input agriculture, no coupling required                                    | STANDS ALONE, GATED       |
| Spirulina               | stand-alone food economics at \$2-4/kg — the ponds need only water and sun                                                    | STANDS ALONE              |
| The board + Mg(OH)&sb2; | coupling-locked: the board's fire retardant IS the brine's magnesium — the pair lives or dies together                        | COUPLING-LOCKED, HONEST   |
| H&sb2;                  | coupling-locked by design: the industrial molecule exists to serve ammonia and the board, never the grid                      | COUPLING-LOCKED, HONEST   |
| The DC                  | optional upside: a separate tenant-backed SPV — removed from the central thesis, exactly as the financing structure demands   | OPTIONAL, EXTERNAL CREDIT |

the strip-down answer: five legs stand alone — water, solar, uranium, terfas, spirulina — and two pairs are coupling-locked by honest physics (the board and the brine's magnesium; the hydrogen and the ammonia). The DC is optional upside on other people's credit. The loop is therefore not a house of cards: it is five independent projects plus two symbiotic pairs, and the page's modular gates are the proof — a module that fails dies alone and the rest of the program does not notice its funeral.

## The credit rating, projected — the way the agencies rate

| FACTOR                  | TODAY                                          | AT FULL PHASE                                                                                                                       |
|-------------------------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Reserves / import cover | 4.5 months today (measured)                    | FX accumulation at full phase: ~\$1.0-1.4B over the 5-year build → 6+ months — the S&P reserves-adequacy factor moves               |
| External position       | CA deficit ~-3.3% of GDP                       | the loop narrows it by ~0.5 pts (279M/yr) → ~-2.5-2.7% — the agencies rate the PATH, not just the level                             |
| Growth                  | ~1.9% today                                    | +0.5 pts from the loop + 1.2 pts from the peace dividend → 3-4% — the strongest single upgrade factor for B-rated sovereigns (Pg40) |
| Fiscal                  | subsidy burden + debt service 26.2% of exports | the substitution legs + the swap's coupon savings + the ESG delta relieve the budget on the same lines                              |

| FACTOR                       | TODAY                                     | AT FULL PHASE                                                                                                     |
|------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Institutional predictability | policy volatility is the chronic negative | the public scoreboard, the gates, the apolitical program — a measurable governance signal the methodology rewards |

**The path: B- → BB- / Ba3 within 3-5 years** (reserves 6+ months, CA ≤ -2.5%, growth ≥ 3% sustained — all inside the measured numbers) and **BB / Ba2 in 5-7 years** with the swap and the nature ledger. Investment grade is NOT claimed: that requires debt/GDP ≤ 60% and a decade of institutional track record — the honest frontier.

the spread arithmetic, stated for what it is: B- to BB- sovereign compression is ~300-500bp; on the refinancing share of the \$40.5B stock that is the state's upside of \$1.2-2.0B/yr — NOT booked in the program's ledger: only the swap's measured \$7.5M/yr and the program-debt ESG delta are credited. The rating is the state's harvest; the program only plants.

## Air quality — the air the children breathe

The board plant caps the phosphogypsum stack dust; the solar leg displaces the grid's gas burn; the biocontrol corridor replaces pesticide drift; and the dew the mist buys scavenges PM from the air (wet deposition). Measured by the PM2.5/NOx/SO2 sensor grid, the dispersion model and the cohort's respiratory KPIs — air quality enters the scoreboard the same way the water does.

## The last inventory — nothing forgotten, nothing hidden

| ITEM                                 | THE GAP IT CLOSSES                                                                           | ITS HOME                                                                                                                                                                                                   | STATUS                             |
|--------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Pond blowdown (the salt bleed)       | the spirulina ponds' own concentrate, when the recycled medium's salts saturate              | routed to the brine line — the pond's salt joins the valorization stream; the water loop closes, the salt loop closes                                                                                      | routing row, added                 |
| Construction carbon & energy payback | the embodied footprint of building the park (PV, concrete, steel) vs the operating recovery  | PV energy payback ~1-2 years (industry standard, DERIVED); the park's embodied carbon is repaid by the displacement + mineralization inside the first operating years — computed in the Phase 0 LCA item   | LCA item in Phase 0                |
| The risk-transfer layer              | drought years, PV underperformance, price shocks — the hazards the gates alone cannot absorb | parametric insurance on the mist corridor's target (rainfall index) and the solar PPA (irradiation index); the degraded cascade already computed is the self-insurance floor                               | insurance, bookable                |
| Nuclear export compliance            | U3O8 leaves the country — the chain of custody the IAEA/NPT regime demands                   | the CNSTN gate already named extends to export controls: buyer compliance, safeguards accounting, transport certification — the Cameco-class compliance chain, cited as a gate, not an afterthought        | compliance chain, gate             |
| The women's leg                      | 54% of women graduates unemployed — who exactly gets the jobs must be designed, not hoped    | the nursery, the workshop, the pond crews and the health workers as women-led cooperatives in the blockade towns — the quota principle already proven in the terfas domestic clause, applied to employment | cooperative structure, contractual |

| ITEM               | THE GAP IT CLOSES                                                                                   | ITS HOME                                                                                                                                                                | STATUS           |
|--------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| The AI's own audit | the dispatcher's code and the twin's models must be auditable or the scoreboard's trust is borrowed | open models, versioned, the prediction-error KPI published, independent code audit before each phase — the nervous system is inspected like everything else it measures | code audit, gate |

# The fauna water ladder — the earwig's drop, as the unit of the whole scale

| SPECIES              | MASS   | WATER / DAY                  | THE NOTE                                                                                                           |
|----------------------|--------|------------------------------|--------------------------------------------------------------------------------------------------------------------|
| The earwig           | 0.1 g  | ~0.5 µL/day (measured class) | one 50 µL drop = 100 earwig-days — the owner's own observation, as the unit of the whole scale                     |
| The wild bee         | 0.1 g  | ~30 µL/day (measured)        | a 10 mL/m <sup>2</sup> dew morning serves ~300 bees per square meter                                               |
| The desert lizard    | 20 g   | ~1.4 mL/day (FMR)            | dew-licking Acanthodactylus — the Saharan micro-fauna's classic water behavior                                     |
| The sandgrouse       | 250 g  | ~10 mL/day (FMR)             | the Sahara's dew-drinker; chicks absorb water through belly feathers — the corridor's fog mornings are its nursery |
| The fennec fox       | 1.2 kg | ~36 mL/day (FMR)             | the desert's smallest canid — water from prey, dew and the corridor's troughs                                      |
| The Egyptian vulture | 2 kg   | ~54 mL/day (FMR)             | globally endangered, breeds in Tunisia — the corridor's carrion-cycling scavenger                                  |
| The Dorcas gazelle   | 17 kg  | ~298 mL/day (FMR)            | endangered; its historic southern steppe is the restoration belt itself                                            |
| The Barbary sheep    | 50 kg  | ~706 mL/day (FMR)            | the mountains' own ungulate — the corridor's upslope fringe is its range                                           |
| The loop's flock     | 60 kg  | ~800 mL/day                  | the agrivoltaic sheep: the mist's 2% is its drinking trough, and its grazing is the land's firebreak               |

one average day's dew on the corridor — 51 m<sup>3</sup>, the same water as one house's annual use — rations across the whole ladder: 10<sup>11</sup> insect-days or 1.7 billion bee-days or a million vulture-days or 170,000 gazelle-days. The species that drink it are the ones Tunisia is losing: the scale that ends at the Dorcas gazelle begins at the earwig's half-microliter — and the mist corridor is the trough for every rung. Allometric spine: daily water = 0.123 x mass<sup>0.8</sup> mL (Nagy, Pg43), cross-checked against measured insect rates.

# The underground boxes — the closed-box theorem, the pond's rival measured head-to-head

| PART        | THE DESIGN                                                                                   | THE NOTE                                                                                            |
|-------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| The chamber | sunken trenches 1.5-2 m below grade — the earth is the insulator: 18-24°C stable, no chiller | underground = thermal inertia; winter nights ride the DC's recovered heat (already in the 33.9 GWh) |

| PART          | THE DESIGN                                                                                                                  | THE NOTE                                                                                                                              |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| The substrate | hydrophilic geo-textile mesh on the walls — the algae's roots are the mesh's pores                                          | the paste scrapes off with a squeegee on a rail: no centrifuges, no flocculants                                                       |
| The roof      | UV-stabilized plastic sheet, sloped — the mist's evaporation condenses on it at night and drips back into the nutrient tank | the closed box: water losses ~10-20% of an open pond's, and the condensate IS the recirculation                                       |
| The misting   | permeate-fed misting bars — our water is low-TDS desal permeate, the ideal anti-clog feedstock                              | misting pressure is a HARD-WIRED safety like the pumps: a 48-72 h outage kills a biofilm, so the interlocks treat it as life-critical |
| The air       | the electrolyzer's oxygen into the chamber air; CO <sub>2</sub> from the DC's air handling                                  | thin-film gas exchange is 3-5x faster than submerged algae — the box breathes better than the pond                                    |

| AXIS          | THE POND                                                                                            | THE BOX                                                                                                                                            | THE READING                                                                              |
|---------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Water         | open pond: ~6 L/m <sup>2</sup> /day evaporates in the desert + blowdown; salinity creeps up forever | closed box: ~0.9 L/m <sup>2</sup> /day class — the 1,000-ha-equivalent spirulina leg could draw ~8,000-10,000 m <sup>3</sup> /d instead of ~32,000 | the box wins where water is the binding constraint — which is everywhere in this country |
| Productivity  | ponds: 5-10 g/m <sup>2</sup> /day (literature-class)                                                | attached/vertical: 20-30 g/m <sup>2</sup> /day claimed in pilots — but COMMERCIAL scale is unproven                                                | the honest number: the claim is 3-5x, the proof is zero until measured                   |
| Harvest       | centrifuge or flocculant — capital + energy                                                         | scrape the paste — a squeegee                                                                                                                      | the box wins on cost per kg harvested                                                    |
| Failure modes | pond: bloom crash, salinity, contamination                                                          | box: misting outage (hours to kill), biofilm sloughing as it thickens (the shading catch), wild-algae invasion                                     | the box has fewer failure modes but the deadly one is FASTER — hence the hard interlocks |

**The gate:** the pilot's head-to-head (1 ha classic ponds as the literature baseline vs 1 ha of the underground boxes): kg/m<sup>2</sup>/yr, m<sup>3</sup> water per kg, kWh per kg, labor hours per kg, paste stability — measured by the same scoreboard, published by the same rule. The winner scales to the 100-ha tier. The box is booked at ZERO in every financial table until the pilot measures it — exactly like the mist and the dew.

the closed box is not the pond's rival — it is the pond's specialist complement, and the terraformation doctrine decides the order. Open water in the desert is the cheapest terraformation instrument on earth: 1,000 ha of ponds evaporating ~60,000 m<sup>3</sup>/day into the local atmosphere is not a loss — it is the loop's SECOND mist system, a permanent blue-water skin that hydrates the sky, feeds the dew, lifts the corridor's green-water recycling (the 10-30% re-precipitation band already booked), and gives the fauna its drinking surfaces. The pond's true output is two goods: paste AND atmospheric water — and the green-water balance prices the second at the recycling band. The boxes earn their place where the atmosphere is already served and thrift or food-safety governs: sealed air against the desert's dust and wild algae, ~85% water saving where the oasis or the aquifer needs the m<sup>3</sup> instead. So the pilot measures the RATIO, not the winner: ponds as the terraformation engine, boxes as the specialist tier — and the desert gets the sky back, one evaporation at a time.

## The domestication gap — the truffle the ancestors ate, and the selection they never applied

| STEP                                | WHAT HAPPENED                                                                                                                                                                                            | THE NOTE                   |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| The ancestors ate it                | desert truffles appear in North African and Roman food records across two millennia — the Berber winter food, gathered after the first rains                                                             | documented, Pg50           |
| The plow never touched it           | the terfas is an obligate mycorrhizal symbiont — it cannot grow without its host shrub — so it never fit the plow-and-seed agriculture that selected wheat and olives; it stayed a gathered wild food    | the biology is the reason  |
| So it was never selected            | no generation applied selection pressure: no varietal improvement, no standardization, no value chain — while France, from the 1970s, domesticated its own truffle into a half-billion-euro industry     | the gap compounds          |
| The wealth went elsewhere           | the world's desert-truffle demand (the Gulf pays premium prices for fresh Tirmania) is served by volatile wild harvests — and the cultivation protocol was invented in Murcia, Spain, for the same genus | the lost option value      |
| The diet lost a protein             | 20-30% protein dry weight, mineral-rich — the ancestors' winter protein diversity was richer than the modern wheat-dependent basin diet                                                                  | the doctor's note, Pg50    |
| The loop finishes the domestication | mycorrhized seedlings (6,000/ha), the gate plot, the quota, the value chain — modern mycology replaces the plow; one decade does what a hundred generations were denied                                  | the program's answer, Pg51 |

the evolutionary point, stated: Tunisia's real lag is not talent — it is SELECTION. Each generation that gathers instead of selects hands the next generation a wider gap, and the product's wealth accumulates to whoever domesticates it first. The terfas is the specimen: eaten for two thousand years, never once selected, finally cultivated — not at home, but in Murcia. The program's deepest leg is therefore not the truffle's revenue: it is the act of DOMESTICATION itself, applied to the whole suite of ancestral assets — the truffle, the medicinal plants, the cactus, the wild bees — and to the brains the country keeps exporting raw. Selection is the oldest technology there is; the loop is the first generation to point it at what the ancestors already knew.

## The oasis, named — the south's date belt, and the aquifer under it

the oasis belt, named: Deglet Nour exports run ~\$250M/yr — one of the country's oldest FX streams, and the oases that grow it sit on the same aquifer the loop protects. The program's answer is already built into its tiers: the inland BWRO + the reuse mesh serve the oasis belt, the household water price falls for the oasis towns like everywhere else, and the scoreboard gains one explicit KPI: **aquifer level recovery**, published quarterly — the south's water account, settled in public. The dates were never an absence; the water under them was.

## The diaspora, named — the country's largest FX source, and the channel that reverses the drain

the diaspora, named: ~\$2.7B/yr of remittances — the country's largest single FX source, sent by the 1.5M Tunisians abroad, many of them the engineers this page has been counting. The program adds two



instruments the bridge never had: a **diaspora bond** against the program's measured KPIs — the people who left buying the future they left — and the **return channel**: the DC/AI tier and the 500-700 operational jobs are the stage the brains were missing. Remittances stay private and untouched; what the program claims is the channel that turns the diaspora's loyalty into the country's capital — the brain-drain's own reversal, priced by the same scoreboard.

The counterfactual, quantified — the truffle against the tree

| AXIS                  | TERFAS                                                                       | OLIVE (THE TUNISIAN DEFAULT)                                                | ALMOND                        |
|-----------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-------------------------------|
| Time to first revenue | 1-3 years (Murcia: first fruit by season 2-3)                                | 5-8 years first fruit, 15-20 to full production                             | 3-4 years first, 8-10 to full |
| Water                 | rained after establishment; years 1-2 only: ~624 m³/ha/yr of drip, then ZERO | 2,000-6,000 m³/ha/yr irrigated — the scarce input                           | 1,000-4,000 m³/ha/yr          |
| Fertilizer            | NONE — the mycorrhizae mine the soil's own phosphorus                        | required, annual                                                            | required, annual              |
| Pesticides            | NONE — the symbiosis has no pest cycle in its climate                        | required (the olive fly)                                                    | required (the almond moth)    |
| Labor                 | harvest season only — the land does the work for 11 months a year            | pruning + harvest + tillage, every year                                     | pruning + harvest + tillage   |
| Capital barrier       | seedlings (6,000/ha) — the only capital; the rural poor can enter            | irrigation system + 20 years of carrying cost — only the capital-rich enter | same trap, shorter            |

the counterfactual, in the register of what was never earned: 940 t/yr at the mature scale — EUR11.3M/yr at the domestic quota, EUR28M at the export mix — multiplied by the fifty years since the domestication technology became available: **EURO.56-1.4B of rural income that never existed**. Not a model error: an inheritance not received.

the Nobel-prize register, stated plainly: the 20-year tree cycle is not a detail of agriculture — it is the intertemporal poverty trap in mathematical form. A rural family without capital cannot wait twenty years, so it plants what returns this year — annuals that deplete the soil — or it plants nothing and migrates. The truffle breaks the trap on every axis at once: 1-3 years to revenue, zero water after establishment, zero fertilizer, zero pesticide, harvest-only labor, and the only capital a seedling costs. The ancestors ate it for two thousand years and never once selected it — the failed agricultural evolution is not a metaphor: it is EURO.5-1.4 billion of rural wealth, five decades of family incomes, and a healthier rural society, all foregone because the plow never reached the symbiont. The loop's terfas leg is therefore not a crop — it is the correction of the oldest mistake in Tunisian agriculture, and the first rural asset whose economics fit the rural poor.

The green-water balance — what the living corridor gives back and filters

| FLOW                                    | THE VOLUME                                   | THE NUMBER                                                                        | THE NOTE                       |
|-----------------------------------------|----------------------------------------------|-----------------------------------------------------------------------------------|--------------------------------|
| Evapotranspiration, returned to the sky | ~17.5M m³/yr of vapor from the restored belt | 0-0% re-precipitates regionally — 1.8-5.2M m³/yr back as rain (van der Ent, Pg44) | the flora multiplies the water |

| FLOW                     | THE VOLUME                                                           | THE NUMBER                                                                                                                                                 | THE NOTE                                           |
|--------------------------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| Infiltration, multiplied | the belt's own ~10M m <sup>3</sup> /yr of rainfall                   | infiltration 15% → 55% with restored cover — ~4.0M m <sup>3</sup> /yr newly captured into the aquifer instead of running off (Pg45)                        | the soil becomes a sponge                          |
| The quality filter       | the root zone + the microbial city filter everything that percolates | 70-90% nitrate removal in the riparian class; the dew-fed food web is the same biological machinery                                                        | the land cleans the water                          |
| The fauna's engineering  | burrows, dung, and the grazing flock                                 | biopores raise infiltration further; manure returns nitrogen to the soil — the animals are the corridor's gardeners (directional, unquantified)            | the species repay the trough                       |
| The insect loop          | dew → insect bodies (~65% water) → predators → decomposition         | every drop an insect drinks becomes a drop in the food web, then a drop returned to the soil on decomposition — the water cycle's trophic plumbing, closed | insects ARE the water cycle, at the smallest scale |

the green-water balance: the mist invests ~0.19M m<sup>3</sup>/yr, and the living corridor cycles ~17.5M m<sup>3</sup>/yr back to the sky — a ~103× amplification of the water's work — plus ~4.0M m<sup>3</sup>/yr newly captured into the aquifer and every drop filtered by the root zone on its way. The honest bounds: the recycling band is the least-certain parameter (booked at its low end), the infiltration multiplier is standard catchment hydrology, and the fauna's engineering is directional. What the mist gives, the land multiplies — that is the bio-recovery the growroom steers toward.

## The rebuild theorem — the mathematics of bringing an ecosystem back

| STATE         | EQUATION                                    | WHAT IT SAYS                                                       |
|---------------|---------------------------------------------|--------------------------------------------------------------------|
| Soil water    | $dW/dt = R + D - L \cdot W - R_2 \cdot W^2$ | rain + dew – evaporation – runoff                                  |
| Biomass       | $dB/dt = J \cdot W \cdot B - M \cdot B$     | water-limited growth – mortality                                   |
| The feedback  | $D = k \cdot ET(B) \cdot e$                 | dew rises with transpiration — the spiral's engine                 |
| The threshold | $R^*$ : the bifurcation                     | below it: bare soil is the only stable state; above it: green wins |

the rebuild theorem, stated as bistability: the SAME land has two effective-rainfall regimes. Bare: crusted soil infiltrates ~15% of its 200 mm rain → ~100 mm effective — at or below the bifurcation threshold  $R^* \sim 100$  mm, where bare soil is the only stable state (Klausmeier, Pg46). Restored: vegetation raises infiltration to ~55% plus the recycling return of its own transpiration — 145-215 mm effective — safely above  $R^*$ , where green is the only stable state. The mist's role is the honest one: its raw dew (0.6 mm/yr) is NOT what crosses the threshold — it is the CATALYST that establishes the vegetation, and the vegetation multiplies the land's own water into the crossing. That is why the subsidy must be persistent and bounded: the system is bistable, so the mist holds it on the green side of the saddle until the vegetation's own recycling takes over — then the mist becomes the guard, not the engine. Testable, as ever: the twin integrates the same equations, the soil grid measures the states, and the prediction error is published.

**The spirulina health claim, at the evidence:** and the health claim, downgraded to the evidence: the 2025 meta-analysis shows positive weight/MUAC effects in malnourished children (Pg41); the 2026 meta-

analysis finds NO significant growth effect (Pg42). The page claims the FOOD, not the cognition — the cohort measures the rest, and the gates decide.

**What this document is:** a concept-level engineering design whose numbers are computed in code from sourced primitives — recomputed, corrected, and applied into the design. Free and public.

**What it is not:** a commissioned state program, a bank-grade feasibility study, an investment offer, or a claim that any of this has been built. Nothing here is advice to any government or any investor. Nothing replaces geological, radiological, geotechnical or EPC-level surveys. Verify every figure and citation before relying on any of it.

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- [Rm12] IAEA / EU BSS: 1 Bq/g (1,000 Bq/kg) exemption threshold for U-238/Ra-226; TENORM classification — <https://www.mdpi.com/2071-1050/17/8/3473...>
- [Rm13] NORM disposal options and costs (\$15-420 per 55-gallon drum) — <https://www.ogj.com/home/article/17230971/norm-disposal-options-costs-vary...>
- [Rm14] China's 85-90% share of global REE separation/refining capacity — <https://www.adlittle.com/sites/default/files/reports/ADL%20The%20rare%20earth%20imperative%202025.pdf...>
- [Rm15] Breakeven economics for non-Chinese REE projects (~\$77/kg NdPr-equivalent; PG-grade requires >\$150/kg) — <https://www.elibrary.imf.org/view/journals/001/2026/072/article-A001-en.xml...>
- [Rm16] Hydrochloric acid prices: EU \$124-150/t (Q4 2025); NA ~\$350/t; China ~\$15/t — <https://chemtradeasia.sg/en/market-insights/hydrochloric-acid-supply-report-2024-2025-buyer-lessons...>
- [Rm17] GCT throughput (~6.5 Mt phosphate rock/yr; >10 Mt/yr PG at peak; 'hundreds of millions of tons' discharged at Gabès) — [http://www.gct.com.tn/fileadmin/user\\_upload/Downloads/Rapport\\_Annuels/GCT-ANNUAL-REPORT-2012-PART1.pdf...](http://www.gct.com.tn/fileadmin/user_upload/Downloads/Rapport_Annuels/GCT-ANNUAL-REPORT-2012-PART1.pdf...)
- [Rw1] Largest SWRO plants: Ras Al-Khair 1,036,000 m<sup>3</sup>/d; Taweelah 909,200 (\$847M-1.2B); Sorek II 672,000; Rabigh 3 600,000 — <http://ide-tech.com/en/project/sorek-b-desalination-plant/...>
- [Rw2] SWRO CAPEX \$800-1,200 per m<sup>3</sup>/day for mega-plants (>100k m<sup>3</sup>/d) — <https://supra-international.com/insights/seawater-reverse-osmosis-swro-desalination-technology-comprehensive-a...>
- [Rw3] LCOW at plant gate: \$0.40-0.60/m<sup>3</sup> (Jubail 3A \$0.41; Sorek II bid \$0.32-0.45); energy share 40-60% of OPEX — <https://pmc.ncbi.nlm.nih.gov/articles/PMC8953854/...>
- [Rw4] Pumping physics: 400 m lift = 1.09 kWh/m<sup>3</sup> at 100% eff; 1.45 at 75%; friction 0.5-1.5 kWh/m<sup>3</sup> over 300 km; realistic 2.0-2.5 kWh/m<sup>3</sup> — [https://www.engineeringtoolbox.com/water-pumping-costs-d\\_1527.html...](https://www.engineeringtoolbox.com/water-pumping-costs-d_1527.html...)
- [Rw5] Jubail-Buraydah IWTP: 587 km, 650,000 m<sup>3</sup>/d, \$2.26B (\$3.85M/km); twin-pipe scale \$5-7M/km — <https://www.meed.com/saudi-arabia-awards-22bn-water-transmission-contract...>
- [Rw6] Delivered inland desalinated cost \$1.10-1.50/m<sup>3</sup> (plant gate + transport + pumping energy) — <https://investriyadh.ai/intelligence/water-scarcity-investment/...>

24. [Rw7] Tunisian water tariffs: SONEDE 0.200-2.310 TND/m<sup>3</sup>; irrigation 0.060-0.160 TND/m<sup>3</sup>; groundwater 0.112-0.162 TND/m<sup>3</sup> — [https://www.sonede.com.tn/tout-savoir-sur-leau/la-facture-deau-methodes-de-paiement/...](https://www.sonede.com.tn/tout-savoir-sur-leau/la-facture-deau-methodes-de-paiement/)
25. [Rw8] International transfer precedents: GMMR \$33B (2.5-6.5 Mm<sup>3</sup>/d, \$0.28-0.62/m<sup>3</sup>, fossil water); Riyadh levelized transport \$0.42/m<sup>3</sup>; <\$0.20/m<sup>3</sup> is the agronomic ceiling — [https://www.mei.edu/publication/whats-next-libyas-great-man-made-river-project/...](https://www.mei.edu/publication/whats-next-libyas-great-man-made-river-project/)
26. [Rw9] 24/7 firming: 3.03 GW @20% CF = 0.606 GW avg (zero margin); 9.15 GWh/night; BESS round-trip 85% → 12-15 GWh; \$110-125/kWh → ~\$1.5B; solar+storage LCOE \$100-130/MWh — <https://www.energy-storage.news/battery-storage-system-prices-continue-to-fall-sharp-ly-bnef-and-ember-reports-...>
27. [Rw10] Tunisia grid: STEG 5.9 GW, ~19 TWh/yr, 95% gas-fired, renewables <5%, industrial tariff 0.352 TND/kWh; budget deficit 5-7% GDP, debt ~83% GDP — <https://www.trade.gov/country-commercial-guides/tunisia-electrical-power-systems-and-renewable-energy...>
28. [Rw11] Agrivoltaics: 20-40% CAPEX premium over utility PV (+\$0.30/Wp); NREL range \$1.20-2.60/Wp — <https://www.mdpi.com/1996-1073/19/1/102...>
29. [Rw12] Tunisia land law: foreign ownership of agricultural land prohibited; 40-year leases only; state-domain conversion is an administrative hurdle — <https://2009-2017.state.gov/e/eb/iftid/2008/101021.htm...>
30. [Rw13] Grid constraints: 3 GW overwhelms local 400 kV; new lines \$1-2.5M/km; renewables <5% of generation; curtailment risk — [https://ireglobal.org/tunisia-focuses-on-grid-expansion-for-integrating-renewable-energy/...](https://ireglobal.org/tunisia-focuses-on-grid-expansion-for-integrating-renewable-energy/)
31. [Rc1] Underground DC precedents: Lefdal Mine (olivine; 200 MW; PUE 1.08-1.15 via 8°C fjord water); SubTropolis limestone (18°C + mechanical cooling); no phosphate-mine DC exists — [https://www.lefdalmine.com/...](https://www.lefdalmine.com/)
32. [Rc2] Gafsa basin: DInSAR-documented subsidence at Moulares; geothermal gradient 25-30°C/km → 23-29°C at 100-300 m depth — [http://www.researchgate.net/publication/251701016\\_Phosphate\\_mine\\_subsidences\\_deduced\\_from\\_differential\\_interf...](http://www.researchgate.net/publication/251701016_Phosphate_mine_subsidences_deduced_from_differential_interf...)
33. [Rc3] Gafsa radiation: 226Ra 16.3-375.1 Bq/kg in soils; phosphate tailings to 7,606 Bq/kg; radon-222 accumulation in enclosed workings; alphas-induced soft errors — [https://www.researchgate.net/publication/380540277\\_Assessment\\_of\\_radiation\\_exposure\\_in\\_the\\_phosphate\\_mining\\_ar...](https://www.researchgate.net/publication/380540277_Assessment_of_radiation_exposure_in_the_phosphate_mining_ar...)
34. [Rc4] ASHRAE TC9.9 Class H1 (5th ed.): 18-22°C recommended, 25°C max for AI accelerators; AI datacenter CAPEX \$20-35M/MW (incl. IT), \$8-15M/MW facility-only — [https://envigilance.com/compliance/ashrae-tc-9-9/...](https://envigilance.com/compliance/ashrae-tc-9-9/)
35. [Rc5] GPU cost: ~100,000 B200-class per 100 MW; \$25-40k each → \$2.5-4B per 100 MW; 300 MW ≈ \$7.5-12B — <https://intuitionlabs.ai/articles/data-center-gpu-pricing-2026...>
36. [Rc6] US BIS export controls 2024-26: UAE 35,000 GPUs (G42/HUMAIN, 'Compute Diplomacy'); Saudi case-by-case 35,000 cap; Tunisia has no bilateral AI-security agreement — <https://www.commerce.gov/news/press-releases/2025/11/statement-uae-and-saudi-chip-exports...>
37. [Rc7] Terfezia cultivation: commercial since 1999-2001 (Murcia, Spain); fruiting at year 2-3; 20-yr studies: average ~376 kg/ha, peak 1,069 kg/ha, drought years 0-2 kg/ha — <https://pmc.ncbi.nlm.nih.gov/articles/PMC8907101/...>
38. [Rc8] Desert-truffle water 200-250 mm/yr; protein 14-22.5% dry weight (the page's '20-27%' sits at the top of the reported range) — <https://www.mdpi.com/2073-4395/11/8/1462...>
39. [Rc9] Terfezia farm-gate €20-60/kg (€200/kg only in extreme scarcity in Gulf markets); Tuber melanosporum €700-1,500/kg retail — different commodities — <https://trufamania.com/desert-truffles.htm...>
40. [Rc10] Mycocluture economics: black truffle payback 10-15 years; Terfezia fruits faster but at one tenth the price — [https://rovirosafarms.com/melanosporum/...](https://rovirosafarms.com/melanosporum/)
41. [Rc11] Tunisia sovereign ratings: Moody's Caa1 (Feb 2025 upgrade, stable); Fitch B- (Sep 2025, affirmed Jan 2026); reserves ~4.5 months import cover; WACC 10-15% — <https://www.fitchratings.com/research/sovereigns/fitch-upgrades-tunisia-to-b-outlook-stable-12-09-2025...>
42. [Rc12] Who finances Tunisia: EIB, AfDB, KfW, World Bank/IFC, EU Global Gateway (17 GW renewable programme); no active IMF program — <https://international-partnerships.ec.europa.eu/policies/global-gateway/17-gw-renewable-energy-programme-tunis...>
43. [Rc13] Empirical IPP record: 500 MW solar awarded 2024/25 (Scatec, Qair, Voltaia 25-yr PPAs at ~\$0.031/kWh); installed renewables ~400 MW; STEG debt >\$1.3B — <https://www.engineeringnews.co.za/article/scatec-awarded-ppas-for-wind-solar-projects-in-tunisia-2026-01-26...>
44. [Rc14] Chain logic: 8 interdependent stages at a generous 70% individual success ⇒ ~5.8% joint probability (0.7<sup>8</sup>) — <https://www.mdpi.com/2073-4395/14/12/3021...>
45. [Rc15] Terfezia clavervyi salt sensitivity: establishes only in near-fresh soils, EC 0.82-0.85 dS/m — [https://jabonline.in/abstract.php?article\\_id=1363&sts=2...](https://jabonline.in/abstract.php?article_id=1363&sts=2...)
46. [Rc16] Terfezia plantation density: ~6,000 mycorrhized plants per hectare — [https://www.researchgate.net/publication/299679495\\_Preparation\\_and\\_Maintenance\\_of\\_Both\\_Man-Planted\\_and\\_Wild\\_Pl...](https://www.researchgate.net/publication/299679495_Preparation_and_Maintenance_of_Both_Man-Planted_and_Wild_Pl...)
47. [Rc17] Saudi Al-Khalidiah 150-ha Tirmania plantation: largely failed (14 kg/ha after soil import; rainfall collapse) — <https://ojs.openagrar.de/index.php/JABFQ/article/download/2184/2570...>
48. [Rc18] Desert-truffle market depth: Mamora (Morocco) wild harvest peaks ~100 t/yr; luxury band breaks past ~1,000-1,500 t/yr of new supply — <https://www.mdpi.com/1999-4907/14/5/952...>
49. [Rc19] Tunisia arid land: ~86,000 ha of state domain earmarked for 40-year agricultural leases (Law 93-120) — <https://documents1.worldbank.org/curated/en/926441468778187741/pdf/multiopage.pdf...>
50. [Ru1] Uranium from wet-process phosphoric acid: Freeport (US) produced 14M lbs U<sub>3</sub>O<sub>8</sub> 1978-99 at >95% DEHPA/TOPO recovery; ~20% of US uranium in the 1980s came from phosphates — <https://www.osti.gov/servlets/purl/1357599...>
51. [Ru2] URANEXT/OCP (Morocco) building the first modern yellowcake-from-phosphoric-acid plant, ~\$100M CAPEX, El Jadida, 2025 — <https://northafricapost.com/87963-uranext-to-build-moroccos-first-yellowcake-uranium-plant-in-el-jadida.html...>

52. [Ru3] WPA uranium SX cost \$59-83/lb U<sub>3</sub>O<sub>8</sub> (IX \$46-76) vs \$80-95/lb 2024-26 spot — <https://repositories.lib.utexas.edu/bitstreams/89897275-85cf-4cd6-a0c5-1d8daa353799/download...>
53. [Ru4] REE from phosphoric acid: PhosAgro pilot; D2EHPA extracts >85% of heavy REEs (Dy, Y, Er) from WPA — <https://www.mdpi.com/2673-4117/7/2/58...>
54. [Ru5] Gafsa phosphate rock uranium content 45-140 ppm (southern basin) — [https://www.researchgate.net/publication/380540277\\_Assessment\\_of\\_radiation\\_exposure\\_in\\_the\\_phosphate\\_mining\\_ar...](https://www.researchgate.net/publication/380540277_Assessment_of_radiation_exposure_in_the_phosphate_mining_ar...)
55. [H1] Guissouma 2017 (Chemosphere): Tunisia tap-water fluoride 0-2.4 mg/L; 25% of population at dental-fluorosis risk, 20% at skeletal-fluorosis risk (WHO limits) — <https://pubmed.ncbi.nlm.nih.gov/28284958/...>
56. [H2] Bouchiba 2026 (Toxics 14(5):438): Mdhilla beneficiation effluent Cd 0.49, Pb 0.71, Cr 3.09 mg/L — exceeding discharge limits; risk driven by Cd, Co, Tl — <https://doi.org/10.3390/toxics14050438...>
57. [H3] Gupta 1996 RCT: fluorosis REVERSED in children (Ca + VitD<sub>3</sub> + ascorbic acid; n=25; water 4.5 ppm F; double-blind) — <https://pubmed.ncbi.nlm.nih.gov/8942013/...>
58. [H4] Ghaemi 2024 (Iran): dental-fluorosis DALY rate up to 981.45 (UI 353.23-1618.40) per 100,000 in a high-F district; Monte Carlo dose-response — <https://pubmed.ncbi.nlm.nih.gov/39003868/...>
59. [H5] Abtahi 2019 (Water Research): age-sex specific DALYs from drinking-water fluoride, Iran national study — <https://pubmed.ncbi.nlm.nih.gov/30953859/...>
60. [H6] Naddafi 2022 (Env Research): burden of disease from heavy metals in drinking water, Iran 2019 — <https://pubmed.ncbi.nlm.nih.gov/34529973/...>
61. [H7] Egner 2014 RCT (n=291): broccoli-sprout beverage 12 wks -> urinary benzene mercapturic acid +61%, acrolein +23% (P<=0.01); GSTT1-positive excrete more — <https://pubmed.ncbi.nlm.nih.gov/24913818/...>
62. [H8] Pabis 2018: zinc supplementation -> >30% reduction of kidney/liver Cd in aged metallothionein-transgenic mice — <https://pubmed.ncbi.nlm.nih.gov/30103140/...>
63. [H9] Zhai 2016: Lactobacillus plantarum CCFM8610 binds gut Cd, protects the intestinal barrier, lowers tissue Cd in mice — <https://pubmed.ncbi.nlm.nih.gov/27208136/...>
64. [H10] Mueller 2018: N95-class respirators >=98% filtration efficiency vs <=44% for any cloth — <https://pubmed.ncbi.nlm.nih.gov/29779694/...>
65. [H11] Allen & Barn 2020: HEPA purifiers substantially cut indoor PM<sub>2.5</sub> and improve subclinical cardiopulmonary health; population benefits often exceed costs — <https://pubmed.ncbi.nlm.nih.gov/33241434/...>
66. [H12] Drieling 2022 RCT (75 asthmatic children): HEPA in bedroom+living room -> poorly-controlled asthma IRR 0.43-0.45; unplanned clinical utilization IRR 0.35 — <https://pubmed.ncbi.nlm.nih.gov/34980119/...>
67. [H13] Spirulina RCT (Cambodia, n=179): 2 g/day x 50 days -> anemia reduction significant (p=0.004); weight-gain trend p=0.07 (do not overclaim growth) — <https://pubmed.ncbi.nlm.nih.gov/36476193/...>
68. [H14] EJAtlas: Gafsa affected population 30,000-70,000; cancer/asthma/infertility reported high; conflict since 2008 — <https://ejatlas.org/conflict/phosphate-mining-in-gafsa...>
69. [H15] NRGi: 'serious dearth of rigorous studies' linking phosphate operations to health outcomes; 7% birth-defect claim is activist-reported, unverified — <https://resourcegovernance.org/articles/uncovering-impacts-phosphate-mining-tunisian-women...>
70. [H16] Geographical 2024: asthma onset age 2 documented (Redeyef); nearest hospital 70 km; doctors prescribe leaving Gafsa — <https://geographical.co.uk/science-environment/gafsa-silent-suffering-the-hidden-cost-of-phosphate-mining-in-...>